Topology-Guided Boundary-Aware Feature Learning for Hyperspectral Image Classification With Dual Graph Networks

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***Abstract*—Spectral variability and boundary ambiguity fundamentally undermine hyperspectral image (HSI) classification by creating inconsistencies between spectral signatures and land-cover categories. Conventional graph convolutional networks (GCNs) fail to resolve these challenges due to indiscriminate message passing across class interfaces. We introduce topology-feature-aware boundary dual GCN (TFAB-DGN), a novel topology-driven framework that integrates magnitude and persistent homology (PH) from algebraic topology into GCNs. Magnitude quantifies spectral–spatial heterogeneity to pinpoint complex boundary regions. PH captures stable topological invariants to recalibrate the adjacency matrix, aligning feature propagation with the intrinsic data structure. This dual-graph architecture hierarchically integrates global spatial–spectral consistencies with local topological invariants. Extensive experiments on four benchmark datasets confirm the superiority of TFAB-DGN over state-of-the-art methods. Ablation studies and visualizations further validate its robustness in resolving boundary ambiguities and establishing a new topology-aware paradigm for HSI analysis.**

***Index Terms*—Boundary enhancement, graph convolutional network (GCN), hyperspectral image (HSI) classification, magnitude, persistent homology (PH).**

I. Introduction

HYPERSPECTRAL images (HSIs) capture the intrinsic spectral characteristics of land covers across hundreds of narrow bands, enabling the precise distinction of subtle material differences [1], [2]. While this rich spectral dimensionality facilitates precise material identification, it simultaneously introduces severe complexity into the data distribution. A critical bottleneck in HSI interpretation is the prevalent spectral variability caused by acquisition conditions and spatial heterogeneity [3], [4]. Consequently, a substantial inconsistency arises between the observed spectral signatures and the actual land cover, creating a nonlinear and confusing feature space. This inconsistency poses a formidable obstacle for traditional classifiers, particularly in scenarios where the boundaries between different land covers become ambiguous.

To disentangle this intricate spatial–spectral distribution, deep learning [3], [5] techniques have emerged as the dominant paradigm for HSI classification. Convolutional neural networks (CNNs) [6], [7], ranging from 1-D spectral to 3-D spatial–spectral architectures [8], [9], [10], have demonstrated remarkable success by learning local spatial and spectral features from regular grid data [11]. More recently, Transformers [12] have further advanced the field by utilizing self-attention mechanisms to capture long-range sequential dependencies that CNNs often struggle to model [13], [14], [15]. Despite their efficacy, both CNNs and Transformers typically operate on regular grid structures (i.e., image patches), which fundamentally limits their ability to characterize the irregular geometric structures inherent in HSIs [16]. Specifically, the fixed square kernels of CNNs and the patch-wise tokenization of Transformers may fail to adaptively aggregate information across complex boundaries, leading to feature blurring in transition areas [17], [18]. This structural mismatch motivates the exploration of graph-based methods, which are naturally better suited for modeling the non-Euclidean relationships among superpixels or distant regions [19].

Graph convolutional networks (GCNs) [20] extend the feature learning of HSIs from regular grids to non-Euclidean spaces [21], [22]. By constructing nodes from specific land-cover regions and edges from their interactions, these networks preserve the spatial consistency of homogeneous areas while capturing long-range dependencies that grid-based models often overlook [23]. While initial attempts treated individual pixels as nodes to model dependencies at the finest scale [24], [25], they were plagued by excessive graph size and prohibitive computational costs [26]. Consequently, superpixel-based methods have gained prominence, transforming the pixel space into structured region-level topological relationships [27], [28], [29]. This shift allows GCNs to more effectively learn the spatial layout and semantic relationships between regions in non-Euclidean space. For example, Cao et al. [30] combined global spatial correlation with

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spectral information through superpixel segmentation and achieved better performance.

Despite their success in modeling regional dependencies, superpixel-based GCNs still grapple with significant challenges in complex boundary regions. The primary bottleneck lies in their heavy reliance on the quality of the initial graphs in which the adjacency matrices are typically constructed using spectral similarity (e.g., Euclidean distance) [31]. However, spectral variability renders these signatures deceptive at class interfaces, particularly in heterogeneous landscapes where spatially adjacent nodes from different classes tend to exhibit high spectral similarity. Such discrepancies lead to an unreliable topological structure, where edges are erroneously established between heterogeneous nodes. Consequently, standard GCNs perform indiscriminative message passing, causing features of boundary nodes to be over-smoothed by their neighbors from different classes [32]. To mitigate this, strategies such as dynamic GCNs (DGCNs) [33], [34] and frequency-aware models [35] have been proposed to refine graph structures or enhance boundary features. Unfortunately, these approaches often lack a robust criterion to explicitly quantify topological complexity, failing to fundamentally regulate information propagation across ambiguous boundaries. Therefore, the key to addressing this dilemma lies in transcending simple spectral metrics and exploiting intrinsic topological properties that can accurately characterize boundary complexity and guide discriminative feature aggregation.

Guided by this perspective, the proposed topology-feature-aware boundary dual GCN (TFAB-DGN) serves as a novel solution that embeds algebraic topology constraints into graph learning. We employ magnitude theory as a high-sensitivity boundary detector, quantifying the intricate spectral–spatial variations to pinpoint potential heterogeneity and key boundary regions. Complementing this local sensitivity, persistent homology (PH) is utilized to extract robust multiscale topological invariants by analyzing connectivity and stable cycle features. By tracking the evolutionary history (birth and death) of these topological features, the framework dynamically reweights the adjacency matrix, effectively pruning misleading interclass edges induced by spectral similarity. This topology-guided optimization ensures clear separation between classes in transition zones, while the dual graph structure facilitates the hierarchical integration of these refined features across multiple scales.

The main contributions are:

1) *Topology-Aware Paradigm (TFAB-DGN):* TFAB-DGN integrates algebraic topology into GCNs to represent HSIs as stable, irregular structures. This dual-graph architecture hierarchically blends global spatial–spectral features with local topological invariants. It resolves the fundamental conflict between feature smoothing and boundary preservation.
2) *Geometric Boundary Localization via Magnitude:* Magnitude serves as a rigorous geometric descriptor to quantify metric space diversity. This mechanism pinpoints spectral–spatial heterogeneity by capturing intrinsic shape properties. It guides the model to focus on hard-to-classify boundary nodes that traditional spectral similarity measures fail to resolve.
3) *Structural Graph Recalibration via PH:* PH extracts multiscale topological invariants such as connected components and cycles to recalibrate the adjacency matrix. This topology-driven refinement filters unreliable edges and ensures discriminative message passing. The resulting framework achieves state-of-the-art performance through intrinsic manifold alignment.

The rest of this article can be summarized as follows. Section II reviews previous related work. Section III elaborates on the architecture of the proposed TFAB-DGN framework. Section IV describes the experiments and analysis. Section V discusses the limitations of the proposed method and directions for future research. Section VI summarizes the work presented in this article.

II. Related Work

A. GCNs-Based Spatial–Spectral Feature Learning

GCNs are a dominant paradigm for modeling HSIs as irregular topological structures. Early improvements like S^2GCN [24] approximate convolutions to mitigate label scarcity. SSGAT [36] introduces attention mechanisms to weigh neighbor importance for discriminative feature learning. Multiscale context is further captured by architectures such as the shifted graph U-Net [37] and the PHHF framework [38]. Despite these developments, standard GCNs remain prone to over-smoothing. Indiscriminate message passing dilutes local details and blurs boundaries between land-cover classes.

Hybrid architectures address these limitations by fusing GCNs with CNNs or Transformers [39]. HCGN [40] and NESSGGCN [18] combine CNN features with graph representations to recover local textures and reduce overfitting. GTFN [41] complements local topological structures with long-range dependencies through Transformers. Most hybrid models still build graphs based on raw spectral similarity. This approach lacks the geometric constraints required to differentiate classes in ambiguous boundary regions. Spectral inconsistency thus continues to limit overall classification performance.

B. Boundary Processing Methods in HSI Classification

Spectral variability and boundary ambiguity create significant challenges for HSI classification. High interclass similarity at class edges often confounds decision-making. CGCNN [42] and BDSNN [43] employ adaptive kernels and deformable networks to match receptive fields with land-cover shapes. Attention-based models refine boundary localization through multiscale features [44]. These approaches improve adaptability through learned correlations. They still struggle with severe spectral inconsistency without a mathematical mechanism for intrinsic geometric stability.

Explicit edge constraints offer an alternative strategy to reinforce boundary awareness. Canny-derived edge maps guide loss computation in [45] to preserve structural details. Statistical node reweighting optimizes graph connectivity in [46] to suppress the influence of ambiguous pixels. These heuristics remain unstable due to the inherent spectral noise of HSIs.

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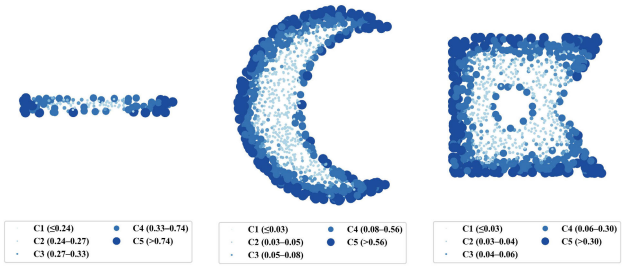


Fig. 1. Boundary awareness of topology-derived magnitude weights. Points in synthetic shapes are sized and colored by their magnitude weight (C1 lowest to C5 highest). The highest-weight points (C4/C5) consistently fall on shape boundaries. This demonstrates the weight's role as a geometry-agnostic boundary indicator, which guides our network to focus on critical regions in HSIs.

Most methods treat boundaries as superficial signal discontinuities rather than intrinsic structural properties [47]. This research gap motivates the use of algebraic topology to characterize geometric stability beyond mere spectral appearance.

C. Algebraic Topology in Image Analysis

1) Magnitude: Algebraic topology, such as magnitude [48], offers a robust framework for analyzing complex data structures by capturing stable shape invariants. This descriptor from category theory serves as a generalization of the Euler characteristic in metric spaces. Foundational studies [49], [50] use magnitude weight vectors to quantify structural heterogeneity. This quantification satisfies the system

$$\zeta_{\mathcal{X}} \cdot w_{\mathcal{X}} = \mathbf{1} \tag{1}$$

where $\zeta_{\mathcal{X}}$ is the similarity matrix between points in a finite metric space $\mathcal{X}$. Similarity is defined as $\zeta_{\mathcal{X}}(i,j) = \exp(-d(x_i,x_j))$, with $d(x_i,x_j)$ representing the distance between two points. The total magnitude is calculated as

$$\mathrm{Mag}(\mathcal{X}) = \mathbf{1}^T \cdot w_{\mathcal{X}}. \tag{2}$$

Internal points have dense neighbor connections and high distance contributions. They must carry smaller weights to satisfy the unit row-sum defined in (1). Boundary points have sparse connectivity and require higher weights to maintain the same equilibrium. These weights consistently peak at the edges of geometric shapes, as shown in Fig. 1. This mechanism establishes magnitude as a robust unsupervised descriptor for boundary identification.

Spectral similarity-based graphs often fail at boundaries due to interclass mimicry. Magnitude directly assesses the geometric diversity of the data space rather than pixel intensity differences. Applying this metric to the Euclidean space of HSI cubes provides a grounded mechanism to extract stable boundary features. This approach effectively circumvents the limitations imposed by spectral variability.

2) Persistent Homology: Captures global structural features by analyzing multiscale topological invariants [51]. It combines simplices such as vertices, edges, and triangles to form a simplicial complex. Fig. 2 illustrates this process using three discrete points. The simplicial complex evolves as the scale

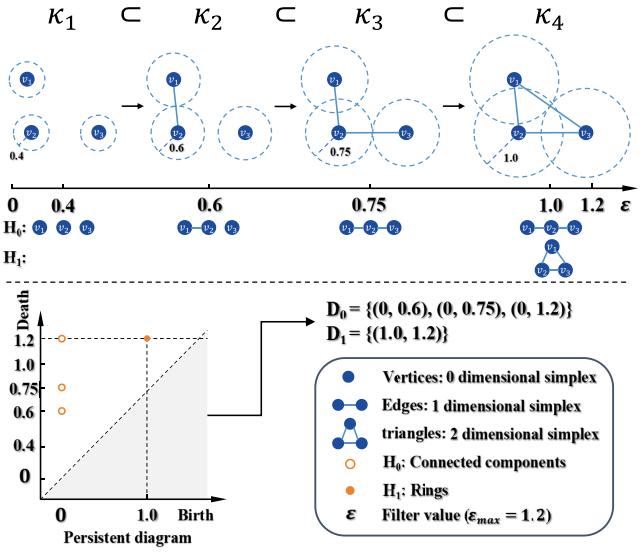


Fig. 2. Illustration of PH via Vietoris–Rips filtration. The upper part shows filtration process: As the scale parameter ϵ increases, the simplicial complex $\mathcal{K}$ evolves from isolated vertices ($\epsilon = 0.4$) to edges and finally a filled triangle ($\epsilon = 1.0$), tracking the birth and death of topological features (H_0: connected components; H_1: rings). The lower part shows PD: The lifespan of each feature is encoded as a point (b,d) in the diagram, where b and d are its birth and death times. The persistence $d-b$ quantifies its topological significance. This framework provides the multiscale topological prior used in our topology-guided feature learning module.

parameter ϵ increases from 0. It sequentially forms isolated vertices (0-D), an edge connecting the closest points (1-D), and a connected component. At $\epsilon = 1.0$, the addition of a third edge fills a triangular face (2-D). This process is known as the Vietoris–Rips filtration, denoted as $\mathcal{K}_1 \subset \mathcal{K}_2 \subset \mathcal{K}_3 \subset \mathcal{K}_4$ [52].

PH quantifies topological features by tracking the birth (b) and death (d) of 0-D (H_0) and 1-D (H_1) homology classes. Each feature is represented by a persistence pair (b,d). In Fig. 2, the pair $(0, 0.6)$ indicates three components at $\epsilon = 0$ where two merges at $\epsilon = 0.6$. The pair $(1.0, 1.2)$ represents a ring born at $\epsilon = 1.0$ that persists until the threshold $\epsilon = 1.2$. These pairs form the persistence diagram (PD) defined as follows:

$$\mathrm{PD}(\mathcal{K}) = \left\{ \mathrm{D}_n = \left\{ \left(b_i^n, d_i^n\right) \right\} \mid b_i^n, d_i^n \in \mathbb{R}, b_i^n < d_i^n, n \in \mathbb{N} \right\} \tag{3}$$

where n denotes n-dimensional topological features and (b_i^n, d_i^n) denote feature i born at b_i^n and dies at d_i^n. Features with a long lifespan $d-b$ indicate significant structures. Short-lived features are treated as noise [53].

Recent studies integrate PH into GCNs to reweight message passing or enhance global context [54], [55]. These models are rarely tailored for the nuances of HSIs. They often assume a reliable initial graph and fail to correct erroneous connections at class boundaries. Spectral variability at HSI boundaries compromises GCNs that rely on misleading spectral similarities. Algebraic topology offers a dual-pronged solution. Magnitude localizes geometric heterogeneity at boundaries. PH extracts robust invariants for complex structural relationships. Integrating these tools allows

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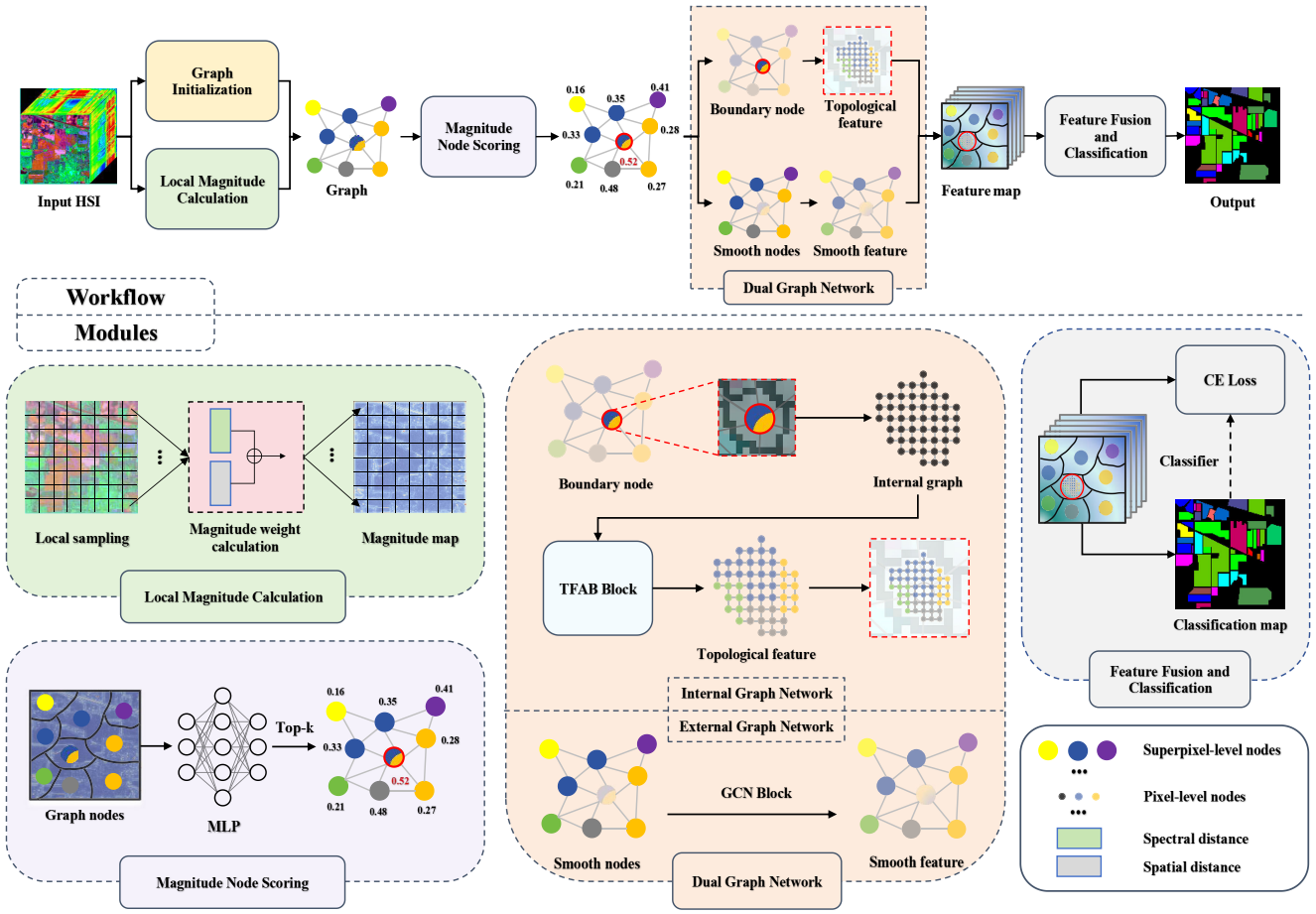


Fig. 3. TFAB-DGN workflow and main modules. The upper portion outlines the workflow through four key modules. These include local magnitude calculation, node scoring, the dual graph network, and feature fusion. The lower portion details the internal structure of each component. Graph initialization follows standard techniques to convert HSIs into graph representations. Section III provides the specific methodology for this process. The TFAB block is shown in Fig. 5.

GCNs to transcend superficial spectral features and recalibrate their message-passing pathways. This synergy motivates our proposed framework to enhance boundary discrimination.

III. Method

The TFAB-DGN architecture is illustrated in Fig. 3 and Algorithm 1. It consists of graph initialization, local Magnitude calculation, node scoring, a dual graph network, feature fusion, and classification. The process begins with spectral dimensionality reduction and SLIC superpixel segmentation to generate the graph structure. Simultaneously, Magnitude computation provides the basis for node scoring. These scored nodes are partitioned into an external graph and an internal graph. The external graph contains smooth nodes, with features extracted through a GCN block. The internal graph comprises boundary nodes, where PH features are captured by the TFAB block. Outputs from both branches are fused and passed through a classifier to produce the final classification map. Each component is detailed below.

A. Graph Initialization

1) Spectral Dimensionality Reduction: The input HSI is defined as $\boldsymbol{X} \in \mathbb{R}^{H\times W\times B}$, where H, W, and B denote height, width, and spectral bands, respectively. Raw data is first channel-normalized according to the following equation:

$$\bar{\boldsymbol{X}} = \frac{\boldsymbol{X} - \min(\boldsymbol{X})}{\max(\boldsymbol{X}) - \min(\boldsymbol{X})}. \tag{4}$$

High-dimensional HSI data requires compression to reduce computational redundancy and noise. We reduce the hundreds of spectral channels to $C(< B)$ key bands while retaining discriminative features. This process involves two iterations of 1×1 convolutions defined as follows:

$$\hat{\boldsymbol{X}}^{l}_{i,j,m} = \sigma\left(\sum_{k} \boldsymbol{W}^{l}_{i,j,m,k} \cdot \mathrm{BN}\left(\bar{\boldsymbol{X}}^{l-1}_{i,j,k}\right) + \mathbf{b}^{l}_{i,j,m,k}\right). \tag{5}$$

Here, i, j denote spatial locations and l represents the layer. $\boldsymbol{W}$ and $\mathbf{b}$ are the weights and biases of the convolution kernels. BN$(\cdot)$ is a batch normalized layer and $\sigma(\cdot)$ denotes the LeakyReLU activation function.

2) SLIC Superpixel Segmentation: The SLIC algorithm [56] segments the HSI into superpixels to balance efficiency with spatial context. Each superpixel acts as a graph node with locally similar spectral features. The input parameter $\delta > 0$ controls the number of superpixels N. Iterative processing

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Algorithm 1 Overall Pipeline of TFAB-DGN

Input: $\boldsymbol{X}$, δ, C, Top-k, L, $\boldsymbol{Y}_{\text{train}}$.
Output: $\boldsymbol{Y}_{\text{pred}}$.

Step1: Graph Initialization
$\hat{\boldsymbol{X}} \leftarrow \text{SpeDimRed}(\boldsymbol{X}, C)$ via Eq.(4), (5);
$\boldsymbol{S}, \boldsymbol{Q}, \boldsymbol{A} \leftarrow \text{SLICSupSeg}(\boldsymbol{X}, \delta)$ via Eq.(6);
$\boldsymbol{H} \leftarrow \text{NodeFeature}(\boldsymbol{Q}, \hat{\boldsymbol{X}})$ via Eq.(7);
Step2: Local Magnitude Calculation
$\boldsymbol{W} \leftarrow$ Algorithm 2;
Step3: Magnitude Node Scoring
for each superpixel node v_i **do**
  $\mathcal{S}(v_i) \leftarrow \text{Scoring}(\boldsymbol{S}, \boldsymbol{H}, \boldsymbol{W})$ via Eq.(12)-(14);
end for
$\text{idx}_{\text{Top-}k} \leftarrow \text{Rank}(\mathcal{S}(v_i), \text{Top-}k)$ via Eq.(15);
Step4: Dual Graph Network
External graph network: GCN Block
$\mathcal{V}_{\text{int}} \leftarrow \text{PartitonNode}(\text{idx}_{\text{Top-}k})$ via Eq.(16);
$\boldsymbol{A}_{\text{ext}}^{(0)}, \boldsymbol{H}_{\text{ext}}^{(0)} \leftarrow \boldsymbol{A}, \boldsymbol{H}$;
for $l = 0$ to $L-1$ **do**
  $\boldsymbol{H}_{\text{ext}}^{(l+1)} \leftarrow \text{GCN}(\boldsymbol{A}_{\text{ext}}^{(l)}, \boldsymbol{H}_{\text{ext}}^{(l)})$ via Eq.(17);
end for
Internal graph network: TFAB Block
for each $v_i \in \mathcal{V}_{\text{int}}$ **do**
  $\mathcal{P}_i \leftarrow$ all pixels belong-ing to v_i;
  $\boldsymbol{A}_{\text{int}}^{i}, \boldsymbol{H}_{\text{int}}^{i} \leftarrow \text{4-Neighborhood}(\mathcal{P}_i)$;
  $\boldsymbol{A}_{\text{re}}^{i} \leftarrow \text{PH}(\mathcal{P}_i, \boldsymbol{A}_{\text{int}}^{i}, \boldsymbol{H}_{\text{int}}^{i})$ via Eq.(18)-(20);
  for $l = 0$ to $L-1$**do**
    $\boldsymbol{H}_{\text{int}}^{i,(l+1)} \leftarrow \text{GCN}(\boldsymbol{A}_{\text{re}}^{i,(l)}, \boldsymbol{H}_{\text{int}}^{i,(l)})$ via Eq.(21);
  end for
end for
Step5: Feature fusion and classification
$\boldsymbol{F}_{\text{final}} \leftarrow \text{Fusion}(\boldsymbol{Q}, \boldsymbol{H}_{\text{ext}}^{\text{out}}, \boldsymbol{H}_{\text{int}}^{\text{out}}, \text{idx}_{\text{Top-}k})$ via Eq.(22)-(24);
$\boldsymbol{Y}_{\text{pred}} \leftarrow \text{Classifier}(\boldsymbol{F}_{\text{final}}, \boldsymbol{Y}_{\text{train}})$ via Eq.(25);
Return $\boldsymbol{Y}_{\text{pred}}$

yields a segmentation matrix $\boldsymbol{S}$, a mapping matrix $\boldsymbol{Q}$, and an adjacency matrix $\boldsymbol{A}$, as summarized as follows:

$$\boldsymbol{S}, \boldsymbol{Q}, \boldsymbol{A} = \text{SLIC}(\boldsymbol{X}, \delta). \tag{6}$$

$\boldsymbol{S} \in \mathbb{Z}^{H \times W}$ contains superpixel labels. $\boldsymbol{Q} \in \mathbb{Z}^{HW \times N}$ defines the pixel-to-superpixel mapping. The adjacency matrix $\boldsymbol{A} \in \mathbb{R}^{N \times N}$ represents connections between spatially adjacent superpixels. Edge weights are based on the Pearson correlation coefficient of spectral features. Node features $\boldsymbol{H} \in \mathbb{R}^{N \times C}$ are calculated as the spectral mean of pixels within each superpixel

$$\boldsymbol{H} = \hat{\boldsymbol{Q}}^{\mathsf{T}} \cdot \text{flatten}\left(\hat{\boldsymbol{X}}\right). \tag{7}$$

The above equation defines the initial node feature matrix using the column-normalized and transposed mapping matrix $\hat{\boldsymbol{Q}}^{\mathsf{T}}$. $\boldsymbol{A}$ and $\boldsymbol{H}$ constitute the initial graph for the network.

B. Local Magnitude Calculation

Given an HSI $\boldsymbol{X} \in \mathbb{R}^{H \times W \times B}$, Algorithm 2 yields a magnitude weight map that reflects the importance of each pixel in its local context.

Algorithm 2 Local Magnitude Calculation for HSIs

Input: $\boldsymbol{X} \in \mathbb{R}^{H \times W \times B}$, window size w_s, parameter α.
Output: Magnitude weight map $\boldsymbol{W} \in \mathbb{R}^{H \times W}$.

Initialize matrix $\boldsymbol{W}$;
Normalize $\boldsymbol{X}$ using min–max per band via Eq.(4);
Create a local window $\boldsymbol{P}$ of size $w_s \times w_s$;
for each pixel $p(x, y)$ **do**
  Determine the local win-dow $\boldsymbol{P}$ of the pixel$p(x, y)$;
  Extract the spectral datawithin the window $\boldsymbol{P}_{x,y}$;
  Take $p(x, y)$ as the cen-tral pixel of the localwindow $\boldsymbol{P}_{x,y}$;
  for each pixel $p(i, j)$ inthe local window $\boldsymbol{P}_{x,y}$**do**
    Calculate spectral distance matrix $\boldsymbol{d}_{\text{spe}}$ via Eq.(8);
    Calculate spatial distance matrix $\boldsymbol{d}_{\text{spa}}$ via Eq.(9);
    Calculate the weighted mixed distance matrix $\boldsymbol{d}_{\text{mix}}$ using parameter α via Eq.(10);
    Assign $D_{\text{mix}}(u, v) = d_{\text{mix}}$;
  end for
  Calculate the similarity matrix $\zeta_{\boldsymbol{P}} = \exp(-D_{\text{mix}})$;
  Calculate the Magnitude weight value $\boldsymbol{W}_{x,y}$ ofthe center pixel $p(x, y)$in window $\boldsymbol{P}_{x,y}$ viaEq. (11);
end for
Normalize $\boldsymbol{W}$ to [0,1];
Return $\boldsymbol{W}$

1) Distance Matrices Computation: Input $\boldsymbol{X}$ undergoes band-wise min–max normalization according to (4). Each pixel $p(x, y)$ defines a local window $\boldsymbol{P}$ of size $w_s \times w_s$. This tunable parameter w_s controls the neighborhood extent. Two distance matrices are computed within this window. Spectral similarity d_{spe} in the following equation captures the cosine distance between spectral vectors $\mathbf{P}_i$ and $\mathbf{P}_j$:

$$d_{\text{spe}}(i, j) = 1 - \frac{\mathbf{P}_i \cdot \mathbf{P}_j}{\|\mathbf{P}_i\| \cdot \|\mathbf{P}_j\|}. \tag{8}$$

Spatial proximity d_{spa} in the following equation represents the Euclidean distance from each pixel (x_i, y_i) to the center (x_c, y_c):

$$d_{\text{spa}}(i, c) = \|(x_i, y_i) - (x_c, y_c)\|. \tag{9}$$

The following equation integrates these metrics into a mixed distance d_{mix} using the importance weight $\alpha \in (0, 1)$:

$$d_{\text{mix}} = \alpha \cdot d_{\text{spe}} + (1 - \alpha) \cdot d_{\text{spa}}. \tag{10}$$

2) Magnitude Weight Calculation: Each local window $\boldsymbol{P}$ constitutes a finite metric space under the metric d_{mix}. The weight vector follows the row-sum constraint defined in (1). An exponential kernel transforms the mixed distance $\boldsymbol{D}_{\text{mix}}$ into the similarity matrix $\zeta_{\boldsymbol{P}} \in \mathbb{R}^{w_s^2 \times w_s^2}$. Magnitude weight $\boldsymbol{W}_{x,y}$ for the central pixel $p(x, y)$ is derived from the inverse similarity matrix via

$$\boldsymbol{W}_{x,y} := \text{Mag}\left(\boldsymbol{P}_{x,y}\right) = \mathbf{1}^T \zeta_{\boldsymbol{P}}^{-1} \mathbf{1}. \tag{11}$$

This pixel-wise calculation produces the global weight map $\boldsymbol{W}$. Min–max normalization scales the final values to the range $[0, 1]$.

As shown in Fig. 4, precomputed magnitude weight maps across four datasets clearly highlight all land-cover boundary contours, effectively capturing image geometric structure

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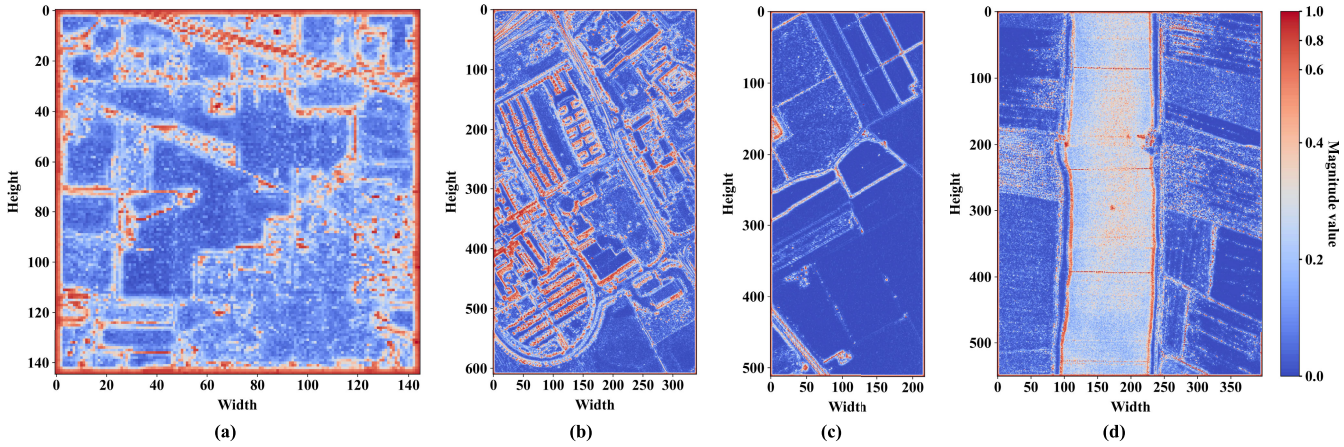


Fig. 4. Precalculated magnitude weight maps on four HSI datasets. Magnitude clearly demarcates boundaries and outlines. (a) IP. (b) PU. (c) SA. (d) LK.

changes beyond spectral differences and providing reliable boundary priors for subsequent scoring. The core of the magnitude weight algorithm lies in the computation of the distance matrix within local windows. To ensure that the distance relationships are not altered, we did not perform dimensionality reduction. This method effectively captures the spectral and spatial correlations in HSIs, providing a crucial basis for boundary detection.

C. Magnitude Node Scoring

1) Scoring Function Definition: Node scoring mechanisms filter key nodes to optimize graph learning in GCNs. This selective approach allows the model to prioritize structural information rather than treating all neighbors equally. Boundary nodes in HSIs often exhibit inconsistencies between spectral signatures and land-cover classes. Traditional GCNs rely on spectral–spatial similarity and fail to capture complex topological relationships at these boundaries. Such limitations lead to ineffective message passing and classification errors. We propose a magnitude-based node scoring mechanism to enhance boundary detection and refine discriminative features.

A graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \boldsymbol{H})$ is constructed from HSIs with superpixel nodes $\mathcal{V} = \{v_1, v_2, \ldots, v_N\}$, edges $\mathcal{E}$, and features $\boldsymbol{H}$. The objective is to learn a scoring function $\mathcal{S} : \mathcal{V} \to \mathbb{R}^+$ to quantify the probability of each node being a boundary node. This function integrates structural characteristics from node features with local topological information from the precomputed magnitude map. The following equation defines the comprehensive score for each node $v_i \in \mathcal{V}$:

$$\mathcal{S}(v_i) = \beta \cdot \mathcal{S}_{\text{mlp}}(v_i) + (1-\beta) \cdot \mathcal{S}_{\text{mag}}(v_i). \tag{12}$$

The learnable fusion parameter $\beta \in (0, 1)$ balances the contribution of both components.

2) Calculation of the Scoring Component: The MLP-based component $\mathcal{S}_{\text{mlp}}(v_i)$ captures structural importance by operating on node features. The following equation defines this calculation:

$$\mathcal{S}_{\text{mlp}}(v_i) = \boldsymbol{W}_2 \cdot \sigma(\boldsymbol{W}_1 \cdot \mathbf{h}_i + \mathbf{b}_1) + \mathbf{b}_2 \tag{13}$$

$\mathbf{h}_i$ is the feature vector of node v_i. $\boldsymbol{W}_1$ and $\boldsymbol{W}_2$ are learnable weights. $\mathbf{b}_1$ and $\mathbf{b}_2$ are bias terms. σ denotes the ReLU activation function. The magnitude-based component $\mathcal{S}_{\text{mag}}(v_i)$ incorporates topological saliency priors by aggregating pixel-level importance within each superpixel according to the following equation:

$$\mathcal{S}_{\text{mag}}(v_i) = \frac{1}{|v_i|} \sum_{k=0}^{|v_i|-1} \boldsymbol{W}(v_{ik}) \tag{14}$$

$|v_i|$ represents the pixel count in node v_i. $\boldsymbol{W}(v_{ik})$ is the magnitude weight of the kth pixel. This component prioritizes nodes in spatially significant regions like land-cover boundaries or areas with distinctive spectral signatures. An importance-based mechanism then selects the Top-k nodes based on these scores to balance boundary focus with computational efficiency. These high-scoring nodes undergo further processing in the network. The following equation expresses the sampling process:

$$\text{idx}^l = \text{rank}\left(\mathcal{S}^l, k\right). \tag{15}$$

The rank$(\cdot)$ function identifies the index idx^l of the selected boundary nodes at layer l. The parameter k controls the number of samples.

D. Dual Graph Network

1) Construction of Dual Graph Networks: Converting HSI data into graph structures involves a fundamental tradeoff. Pixel-level methods significantly increase graph scale and resource consumption. Superpixel-based approaches alleviate this scale issue but sacrifice internal details. We propose an internal–external dual graph structure to address this limitation. This framework classifies nodes into two distinct types. It enables differentiated feature learning tailored to the specific characteristics of boundary and smooth regions.

2) External Graph Network: Given the scored graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \boldsymbol{H})$, we partition the node set into two mutually exclusive subsets based on the node scores $\mathcal{S}(v_i)$

$$\mathcal{V} = \mathcal{V}_{\text{int}} \cup \mathcal{V}_{\text{ext}} \tag{16}$$

where $\mathcal{V}_{\text{int}} = \{v_i \mid i \in \text{idx}^l\}$ contains nodes from the boundary regions with higher scores, and $\mathcal{V}_{\text{ext}} = \{v_i \mid i \notin \text{idx}^l\}$ contains nodes from the smooth regions with lower scores. The

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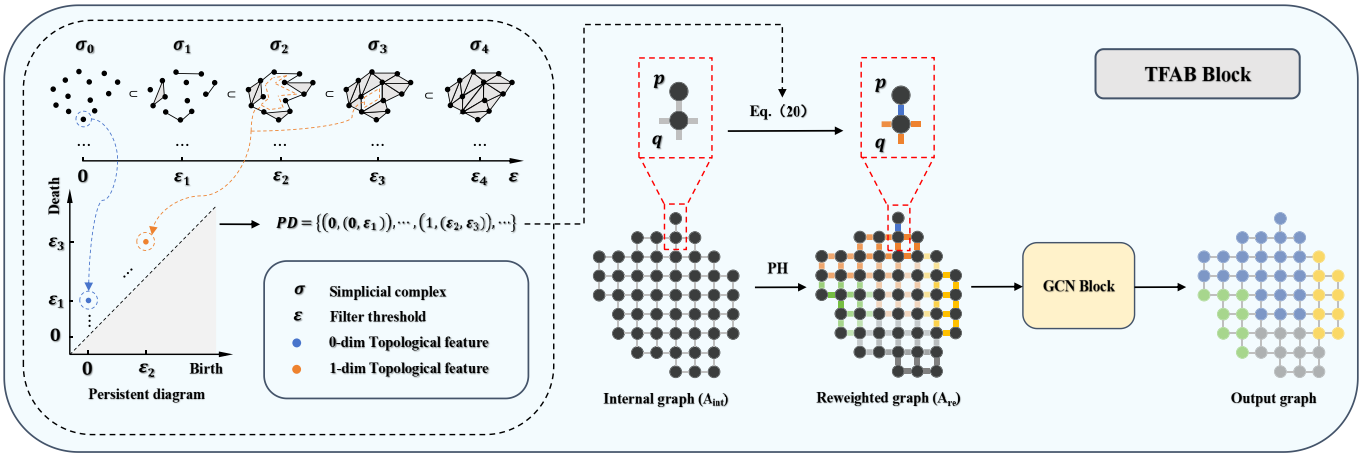


Fig. 5. Illustration of the TFAB block. The module first identifies boundary nodes to form an internal graph. It then employs PH to track the birth and death of 0-D and 1-D topological features across varying filtration scales (ϵ). These topological priors are used to reweight edges—prioritizing cycles and high-persistence connections—according to (20). The reweighted graph is then processed by a GCN to capture boundary-sensitive features.

external graph $\mathcal{G}_{\text{ext}} = (\mathcal{V}_{\text{ext}}, \mathcal{E}_{\text{ext}}, \boldsymbol{H}_{\text{ext}})$ primarily contains nodes from regions with relatively uniform spectral–spatial features. We employ a standard graph convolutional layer for feature extraction, with its single-layer operation typically expressed as

$$\boldsymbol{H}_{\text{ext}}^{(l+1)} = \sigma\left(\tilde{\boldsymbol{D}}_{\text{ext}}^{-\frac{1}{2}} \tilde{\boldsymbol{A}}_{\text{ext}} \tilde{\boldsymbol{D}}_{\text{ext}}^{-\frac{1}{2}} \boldsymbol{H}_{\text{ext}}^{l} \boldsymbol{W}_{\text{ext}}^{l}\right) \tag{17}$$

where $\tilde{\boldsymbol{A}}_{\text{ext}} = \boldsymbol{A}_{\text{ext}} + \boldsymbol{I}$ represents the self-loop adjacency matrix of $\boldsymbol{A}_{\text{ext}} \in \mathbb{R}^{N\times N}$ calculated using (6) after adding self-loops, $\tilde{\boldsymbol{D}}_{\text{ext}} = \text{diag}(\tilde{\boldsymbol{A}}_{\text{ext}} \cdot \boldsymbol{1})$ is its degree matrix, $\boldsymbol{H}_{\text{ext}}^{l} \in \mathbb{R}^{N\times C}$ represents the node feature matrix of the lth layer of the external representation calculated using (7), $\boldsymbol{W}_{\text{ext}}^{l}$ is the learnable weight matrix, and σ is the activation function.

3) Internal Graph Network: The internal graph is designed to handle the complex topological structure of boundary regions, and the TFAB block construction process is illustrated in Fig. 5. Boundary nodes are represented using pixel-level graphs, and topological features are extracted through PH to guide information propagation. For each selected boundary node $v_i \in \mathcal{V}_{\text{int}}$, we extract all pixels belonging to node v_i and represent this set of pixels as $\mathcal{P}_i$. To preserve the local spatial relationships in the boundary region, we use the four-neighborhood to construct the internal graph adjacency matrix $\boldsymbol{A}_{\text{int}}^{l} \in [0,1]^{|\mathcal{P}_i|\times|\mathcal{P}_i|}$ at the lth layer, where $|\mathcal{P}_i|$ denotes the number of pixels it contains.

However, the edge weights of the adjacency matrix $\boldsymbol{A}_{\text{int}}^{l}$ are all one. To enhance the discriminative capability of the message passing process, it is necessary to reassign the edge weights. In addition, traditional GCNs may result in incorrect information aggregation due to the highly similar spectral characteristics of the boundaries. To overcome these challenges, we propose integrating topological information into the edge weights and using it to weight the adjacency matrix.

Specifically, for the unweighted adjacency matrix $\boldsymbol{A}_{\text{int}}^{l} \in [0,1]^{|\mathcal{P}|\times|\mathcal{P}|}$ and node feature matrix $\boldsymbol{H}_{\text{int}}^{l} \in \mathbb{R}^{|\mathcal{P}|\times C}$ of the internal graph, we first compute the Euclidean distance matrix, defined as

$$\boldsymbol{D}_{\text{euc}}^{l}(p,q) = \|\boldsymbol{H}_{p}^{l} - \boldsymbol{H}_{q}^{l}\|_2 \tag{18}$$

where p and q represent pixels in the internal graph and $\boldsymbol{H}_{p}^{l}$ and $\boldsymbol{H}_{q}^{l}$ are the feature vectors of pixels p and q at the lth layer, respectively. Then, based on the distance matrix between nodes, the simplicial complex is constructed to capture the topological relationships between nodes [57]. The simplicial complex is a set that contains all possible simplices, where the dimension of each simplex is determined by the distance between nodes. A 0-D simplex represents a single node, and if the distance between two nodes is less than or equal to a threshold ϵ, a 1-D simplex forms an edge, and so on. Therefore, given the pixel node set $\mathcal{P} = \{v_1, v_2, \ldots, v_n\}$ and distance matrix $\boldsymbol{D}_{\text{euc}} \in \mathbb{R}^{n\times n}$ of an internal graph, the construction process of all simplicial complexes can be represented as

$$\mathcal{K}(\boldsymbol{D}_{\text{euc}}, \epsilon) = \left\{\sigma \subseteq \mathcal{P} \mid \max_{p,q\in\sigma} \boldsymbol{D}_{\text{euc}}(p,q) \le \epsilon\right\} \tag{19}$$

where $\sigma = \{v_{i_0}, v_{i_1}, \ldots, v_{i_k}\}$ represents the set of simplicial complexes, and ϵ is the distance threshold that controls the scale of the complex. This abstract topological structure, which is constructed based on the distances between discrete nodes, is referred to as a Vietoris–Rips complex [52].

Next, PH extracts the topological features of the Vietoris–Rips complex, which can be represented by the PD via (3). Due to the high computational cost and unstable results in high-dimensional cases, typically only 0-D and 1-D topological features are computed, so n takes 0 and 1. The PD stores this persistent global topological information [58]. To inject this topological information into the graph convolution, we reweight the adjacency matrix $\boldsymbol{A}_{\text{int}}$. This can be defined as

$$\boldsymbol{A}_{\text{re}}(p,q) = \begin{cases} d_i^1 - b_i^1, & \exists i\in\mathbb{I}_1, \sigma_k\in\mathcal{R},\ \text{s.t.}\ (v_p,v_q)\in\text{edge}(\sigma_k) \\ & \wedge\ (b_i^1, d_i^1)\in\text{D}_1\ \wedge\ \boldsymbol{A}_{\text{int}}(p,q)=1 \\ \max\left(d_p^0, d_q^0\right), & \text{otherwise} \end{cases} \tag{20}$$

where $\text{edge}(\sigma_k) = \{(v_i, v_j) \mid v_i, v_j \in \sigma_k\}$ represents the set of all 1-D simplices in the simplicial complex σ_k. When the first condition in (20) is satisfied, it indicates that the edge (v_p, v_q) in $\boldsymbol{A}_{\text{int}}$ is also an edge of a 1-D cycle (1-D topological feature) in the kth simplicial complex. In the above case, the weight is defined as the persistence value of the 1-D topological feature.

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In other cases, the persistent value of the edge is taken as the maximum of the 0-D persistent values of the two nodes p and q, since the birth of 0-D simplices is always 0, and thus the persistent value equals the death value.

After reweighting the adjacency matrix, we perform graph convolution on the internal graph to extract topologically aware features. The TFAB block employs a GCN layer that integrates the PH-weighted adjacency matrix. The convolution operation for the internal graph of the lth layer is formulated as

$$\boldsymbol{H}_{\text{int}}^{(l+1)} = \sigma\left(\tilde{\boldsymbol{D}}_{\text{int}}^{-\frac{1}{2}} \tilde{\boldsymbol{A}}_{\text{re}}^{l} \tilde{\boldsymbol{D}}_{\text{int}}^{-\frac{1}{2}} \boldsymbol{H}_{\text{int}}^{l} \boldsymbol{W}_{\text{int}}^{l}\right) \tag{21}$$

where $\tilde{\boldsymbol{A}}_{\text{re}}^{l} = \boldsymbol{A}_{\text{re}}^{l} + \boldsymbol{I}$ is the self-loop weighted adjacency matrix, $\tilde{\boldsymbol{D}}_{\text{int}} = \text{diag}(\tilde{\boldsymbol{A}}_{\text{re}} \cdot \boldsymbol{1})$ is its degree matrix, $\boldsymbol{W}_{\text{int}}^{l}$ is a learnable weight matrix, and σ is the activation function. Edges that constitute stable topological structures are assigned higher weights, enabling feature propagation to follow the underlying topological structure.

As the number of Top-k nodes increases, PH in the internal graph faces the challenge of transitioning from handling complex boundaries to addressing noise. Berens et al. [59] pointed out that in datasets with low intrinsic dimensions (e.g., HSI), existing in a higher-dimensional space makes PH particularly sensitive to noise, which hinders the ability to detect the correct topological structure. Furthermore, the computational complexity of PH grows exponentially with the number of nodes, resulting in a limited ability to process large-scale graphs [60]. Thus, employing TFAB to handle a small number of local boundary nodes while using GCNs to process the remaining smooth nodes ensures theoretical global consistency.

E. Feature Fusion and Classification

1) Feature Fusion: The outputs from the external graph and internal graph branches are fused to combine smooth region features and boundary region topological features. Since the external graph operates at the superpixel level, we assign each superpixel-level feature back to the pixel level through the mapping matrix $\boldsymbol{Q}$, which can be expressed as

$$\boldsymbol{F}_{\text{ext}}^{\text{out}} = \text{Reshape}\left(\boldsymbol{Q}\boldsymbol{H}_{\text{ext}}^{\text{out}}\right) \tag{22}$$

where $\boldsymbol{H}_{\text{ext}}^{\text{out}}$ represents the output node feature matrix of the external graph and Reshape$(\cdot)$ denotes the reshaping to the original data space dimension. Based on the segmentation matrix $\boldsymbol{S}$ and the stored boundary node indices idx, the output node feature matrix $\boldsymbol{H}_{\text{int}}^{\text{out}}$ of the internal graph is fused with the features $\boldsymbol{F}_{\text{ext}}^{\text{out}}$ of the external graph. This process can be expressed as

$$\boldsymbol{F}_{\text{out}}(p) = \begin{cases} \boldsymbol{H}_{\text{int}}^{\text{out}}(p), & \text{idx}_p \in \text{idx} \\ \boldsymbol{F}_{\text{ext}}^{\text{out}}(p), & \text{idx}_p \notin \text{idx} \end{cases} \tag{23}$$

where idx_p represents whether the index of the superpixel node to which pixel p belongs is included in the Top-k index set idx. To integrate multiscale information, learnable weights are assigned to the features outputted by each scale, and the weighted sum is calculated, which can be expressed as

$$\boldsymbol{F}_{\text{final}} = \sum_{l=1}^{L} \alpha_l \cdot \boldsymbol{F}_{\text{out}}^{l} \tag{24}$$

where α_l is the learnable fusion weight for the output features of the lth layer.

2) Classification: To further enhance the discriminative power of the features, the integrated multiscale features $\boldsymbol{F}_{\text{final}}$ are fed into a 2-D convolutional layer. The final output is obtained through a linear layer, followed by a Softmax layer to predict the class probabilities for each pixel. And the classical cross-entropy error is adopted as the HSI classification loss $\mathcal{L}$, which is defined as

$$\mathcal{L} = -\sum_{i=1}^{T}\sum_{j=1}^{N} \boldsymbol{Y}_{ij} \log\left(\hat{\boldsymbol{Y}}_{ij}\right) \tag{25}$$

where T and N are the number of training examples and classes, respectively, $\log(\cdot)$ is the natural logarithm, and $\boldsymbol{Y}_{ij}$ and $\hat{\boldsymbol{Y}}_{ij}$ represent the ground-truth label and prediction matrix, respectively.

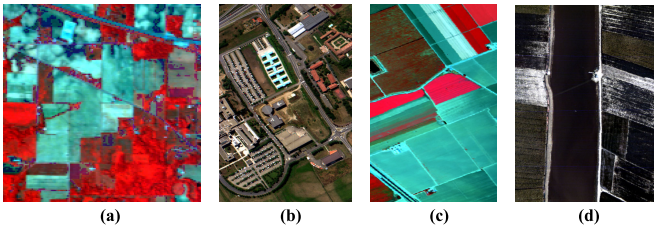


Fig. 6. False-color images of the four datasets. (a) IP. (b) PU. (c) SA. (d) LK.

IV. Experimental Analysis

This section verifies the effectiveness of TFAB-DGN in HSI classification and employs ablation studies along with parameter analysis to reveal the critical role of topological tools in the proposed algorithm. To validate the classification performance, quantitative comparisons of multiple metrics are conducted against mainstream classification algorithms on four benchmark HSI datasets. Meanwhile, visualized classification results are provided to illustrate potential reasons for performance differences. To elucidate the key contribution of topological tools in the TFAB-DGN algorithm, ablation experiments are carried out, demonstrating the significant impact of two types of topological indicators on classification accuracy. Furthermore, additional experiments are performed to further clarify the role of the topological indicator. Finally, the computational complexity of the TFAB-DGN algorithm is discussed.

A. Dataset Description and Experimental Settings

Method evaluation uses four benchmarks summarized in Table I: Indian Pines (IP), Pavia University (PU), Salinas (SA), and WHU-Hi-LongKou (LK) [61]. Fig. 6(a)–(d) shows their respective false-color images. IP features 145 × 145 pixels at 20-m resolution, covering a 0.4–2.5-μm range with

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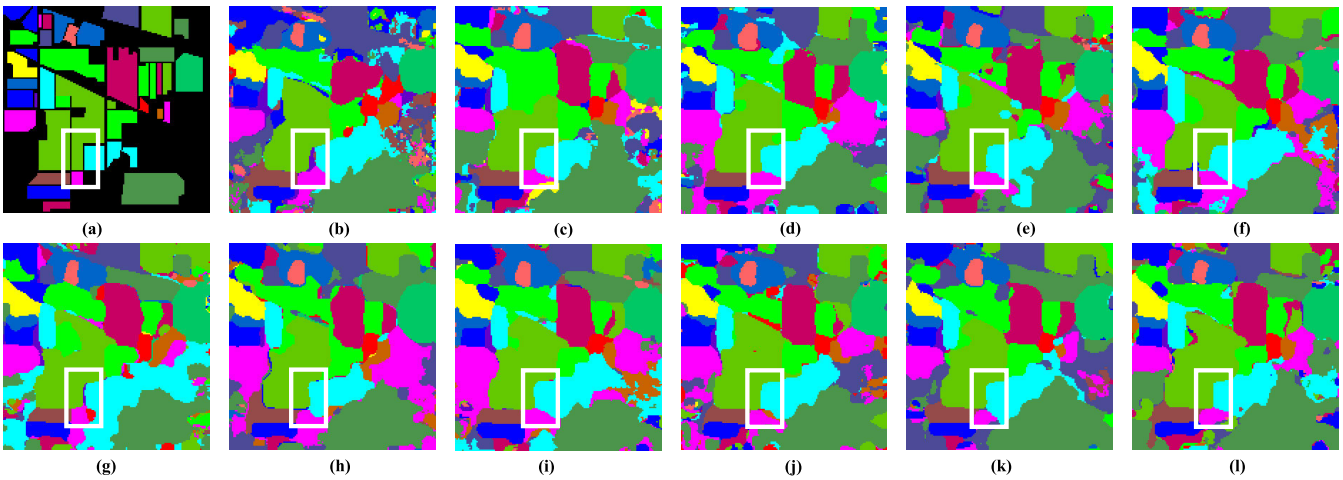


Fig. 7. Classification maps of different methods on the IP dataset. (a) Ground truth. (b) 3DCNN. (c) DBMST. (d) SPGCN. (e) OGU. (f) SPGAU. (g) LPDGCN. (h) GiGCN. (i) NESSGGCN. (j) HCGN. (k) TFAP-GUN. (l) Proposed.

TABLE I
TRAIN AND TEST NUMBER OF SAMPLES PER CLASS FOR ALL FOUR DATASETS

Dataset		IP		PU		SA		LK	
Class No.	Color	Train	Test	Train	Test	Train	Test	Train	Test
1		30	16	30	6601	30	1979	30	34481
2		30	1398	30	18619	30	3696	30	8344
3		30	800	30	2069	30	1946	30	3001
4		30	207	30	3034	30	1364	30	63182
5		30	453	30	1315	30	2648	30	4121
6		30	700	30	4999	30	3929	30	11824
7		15	13	30	1300	30	3549	30	67026
8		30	448	30	3652	30	11241	30	7094
9		15	5	30	917	30	6173	30	5199
10		30	942			30	324		
11		30	2425			30	1038		
12		30	563			30	1897		
13		30	175			30	886		
14		30	1235			30	1040		
15		30	356			30	7238		
16		30	63			30	1777		
Total		450	9799	270	42506	480	53649	450	204272

200 processed bands across 16 categories. PU comprises 610 × 340 pixels at 1.3-m resolution, providing 103 spectral bands (0.43–0.86 μm) for 9 classes. SA contains 54129 samples over 16 categories, with a 512 × 217 resolution at 3.7 m (0.2–2.4 μm) and 204 spectral bands. LK offers a fine 0.463-m spatial resolution across 550 × 400 pixels, including nine surface types and 270 bands within 0.4–1.0 μm.

We evaluate TFAB-DGN against state-of-the-art methods: 3DCNN [10], DBMST [62], and several GCN-based models [18], [28], [29], [37], [40], [63], [64], [65]. Each experiment is repeated 10 times to average the results from random sampling and initialization. Training data includes 30 samples per class, or 15 if fewer exist. The remaining samples form an independent testing set.

All models are implemented in PyTorch on a single system featuring an Intel i9-13900KF CPU and an NVIDIA GTX TITAN X GPU. Hyperparameters include two multiscale hidden layers and a 0.0005 learning rate with the Adam optimizer. Training runs for 200 epochs on IP and 150 epochs on the other three datasets.

Performance metrics include per-class accuracy, OA, AA, Kappa, and $F1$-score. Visualized classification maps supplement these quantitative results. Model complexity is measured by training time, testing time, and parameter counts. Detailed hyperparameter sensitivities are discussed in Section IV-E.

B. Classification Performance

TFAB-DGN is evaluated through quantitative metrics and qualitative visualizations. Tables II–V show that our model achieves optimal OA, AA, Kappa, and $F1$ values across all datasets. It attains the highest classification accuracy on multiple categories, including seven 100% scores on the IP dataset. Compared to GiGCN on the PU dataset, TFAB-DGN improves OA, AA, Kappa, and $F1$ by 3.38%, 2.00%, 6.19%, and 4.77%, respectively. Our model also outperforms NESSGGCN and HCGN on the SA and LK datasets. Smaller standard deviations across all metrics further demonstrate superior model stability.

Figs. 7–10 illustrate classification maps that closely match actual land-cover distributions. White boxes highlight regions where boundary predictions differ most between TFAB-DGN and other methods. Our approach produces accurate contours in IP and LK maps while reducing the jagged or blurry edges common in other models. It maintains clear boundaries in PU and SA datasets despite significant background interference. These results validate the ability of TFAB-DGN to capture HSI features and distinguish category boundaries effectively.

C. Ablation Studies

Systematic ablation experiments validate magnitude node scoring (MNS) and the TFAB block. "W/O MNS" removes magnitude topological information from node scoring. "W/O TFAB" eliminates PH-based weighting in the internal graph. "W/O MNS&TFAB" represents a traditional GCN using MLP scoring and unweighted convolutions.

Boundary-overall accuracy (B-OA) is introduced to address HSI boundary classification challenges [66]. Material

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TABLE II

ACCURACY COMPARISONS ON THE IP DATASET WITH 30 TRAINING SAMPLES. BOLD: BEST; UNDERLINE: SECOND-BEST

Methods	CNN	Transformer	GCN								
Class	3DCNN	DBMST	SPGCN	OGU	SPGAU	LPDGCN	GiGCN	NESSGGCN	HCGN	TFAP-GUN	Proposed
1	96.87±5.03	95.45±4.65	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	99.73±0.12	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
2	70.90±4.85	91.03±3.11	94.06±2.16	91.84±2.63	96.91±2.64	86.19±3.73	80.96±3.14	**97.56±1.26**	97.06±1.79	88.68±3.00	88.05±5.51
3	65.70±6.09	91.84±1.43	92.75±3.13	93.12±3.14	**96.37±2.16**	93.00±2.05	91.77±1.39	94.62±2.59	92.37±3.12	94.83±2.58	88.63±4.36
4	84.73±4.10	94.87±5.72	92.75±4.15	96.95±0.28	91.30±3.59	99.51±0.58	80.77±1.93	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
5	86.35±8.02	94.62±0.75	95.80±2.56	94.77±2.36	97.13±1.67	91.61±2.63	90.75±1.41	**98.89±1.23**	96.18±2.41	94.02±3.26	97.57±2.63
6	92.68±4.88	98.43±0.32	90.85±5.16	95.17±0.73	98.15±0.79	98.57±1.46	94.69±1.25	96.71±2.16	97.15±1.34	98.15±1.14	**99.86±0.24**
7	96.92±3.76	92.58±4.08	**100.00±0.00**	97.59±1.18	99.16±0.89	92.30±2.30	74.31±2.93	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
8	96.80±1.99	97.87±0.09	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	99.32±0.78	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
9	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	85.16±2.13	96.92±2.16	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
10	78.57±5.26	91.73±6.65	91.71±3.56	91.02±2.76	92.78±2.19	93.16±2.91	88.57±1.92	86.26±5.13	88.11±3.79	92.63±3.37	**97.98±2.35**
11	64.39±5.27	91.23±1.05	87.54±7.13	90.02±1.80	86.55±5.67	90.42±3.55	92.75±1.45	**96.09±2.45**	90.39±3.19	91.78±2.91	94.23±4.29
12	77.94±5.59	89.31±8.73	**93.96±3.16**	92.91±4.39	91.11±3.26	92.02±3.11	90.81±1.48	92.16±3.49	90.05±3.89	91.83±2.42	93.25±3.21
13	99.54±0.66	99.19±0.16	98.15±1.13	**100.00±0.00**	**100.00±0.00**	99.42±0.51	98.98±0.34	99.42±0.35	99.12±0.82	99.64±0.37	99.43±0.22
14	86.38±12.11	96.02±1.26	**99.83±0.26**	95.76±2.85	95.62±3.59	97.57±3.59	97.57±0.84	99.12±0.79	99.34±0.64	97.36±2.72	99.43±0.64
15	73.51±8.23	92.98±5.26	99.15±0.81	99.85±0.51	99.51±0.43	99.66±0.83	93.87±1.17	**100.00±0.00**	**100.00±0.00**	99.83±0.25	**100.00±0.00**
16	98.73±1.18	94.82±0.98	96.82±2.16	**100.00±0.00**	99.15±0.81	99.52±0.72	98.88±0.41	**100.00±0.00**	99.59±0.19	98.89±2.01	**100.00±0.00**
OA (%)	76.58±2.60	92.64±2.53	93.24±1.61	93.84±0.55	93.56±1.26	94.50±0.98	92.86±1.12	95.07±1.26	94.44±1.56	93.99±0.81	**95.16±0.52**
AA (%)	85.62±1.33	93.49±1.13	95.95±1.43	95.31±0.29	96.73±1.78	94.16±0.35	91.59±0.95	97.09±0.75	97.12±1.16	96.73±0.34	**97.39±0.82**
Kappa (%)	73.59±2.79	91.12±1.98	92.31±2.63	92.38±0.62	93.00±2.59	93.88±1.11	90.74±1.23	94.28±1.61	93.81±1.94	93.13±0.91	**94.46±1.02**
F1 (%)	71.56±2.22	89.86±1.48	89.63±2.98	87.59±2.01	92.67±2.37	90.12±1.06	88.59±2.63	93.07±0.67	92.97±0.98	89.35±1.73	**93.17±1.71**

TABLE III

ACCURACY COMPARISONS ON THE PU DATASET WITH 30 TRAINING SAMPLES. BOLD: BEST; UNDERLINE: SECOND-BEST

Methods	CNN	Transformer	GCN								
Class	3DCNN	DBMST	SPGCN	OGU	SPGAU	LPDGCN	GiGCN	NESSGGCN	HCGN	TFAP-GUN	Proposed
1	79.42±6.58	97.35±1.46	95.74±0.93	91.15±2.15	99.21±0.75	89.86±3.61	**99.31±0.71**	93.30±2.25	97.53±2.34	93.46±4.81	97.04±1.02
2	69.36±6.11	98.21±0.89	96.87±1.59	94.64±3.89	97.76±1.43	93.28±3.08	82.91±8.24	97.67±1.54	**98.93±1.19**	91.56±5.29	97.16±1.18
3	84.73±4.92	92.76±2.59	82.39±2.69	98.67±2.89	94.30±1.41	98.46±2.19	86.37±2.63	88.21±3.32	95.92±1.95	95.79±5.15	**99.82±0.65**
4	92.81±1.49	**97.47±1.03**	96.38±1.66	89.78±3.18	90.28±2.11	95.14±2.92	96.52±1.42	93.67±2.16	92.36±2.94	92.53±4.78	94.03±1.56
5	99.61±0.42	98.41±0.55	99.04±1.07	99.94±0.09	99.13±0.79	99.38±0.84	99.92±0.08	99.25±0.25	98.92±1.06	99.11±0.98	**100.00±0.00**
6	78.12±4.84	99.03±0.33	95.56±1.28	99.69±0.29	95.78±0.67	99.83±0.24	**99.94±0.06**	93.36±3.69	98.83±1.03	99.22±1.28	99.63±0.43
7	89.48±5.88	98.03±0.87	97.58±1.07	**100.00±0.00**	98.89±0.99	97.49±4.88	92.64±3.16	96.58±2.58	97.32±1.48	93.19±3.53	**100.00±0.00**
8	85.76±3.38	91.74±3.24	92.56±2.24	97.59±1.70	96.26±1.35	91.92±4.65	94.13±2.54	89.03±2.16	97.55±1.79	95.32±3.41	**98.09±1.23**
9	99.45±0.49	97.43±1.11	98.02±0.51	95.13±2.41	95.03±0.94	96.56±1.09	**99.46±0.16**	96.16±2.65	97.65±3.64	97.13±1.41	99.09±1.67
OA (%)	77.99±6.05	96.69±1.26	95.87±0.40	95.66±1.64	96.67±1.24	96.45±1.41	94.27±2.12	97.12±0.54	97.45±1.49	95.14±1.46	**97.65±0.39**
AA (%)	86.53±2.14	95.13±1.23	95.24±0.63	95.98±0.61	95.21±1.29	96.02±1.28	96.35±2.14	96.53±2.64	97.28±1.67	96.03±1.08	**98.35±0.24**
Kappa (%)	72.55±6.46	95.94±1.03	94.46±1.54	94.27±2.07	94.89±1.28	95.89±1.78	90.63±3.44	95.64±2.18	96.26±1.29	93.56±3.07	**96.82±0.58**
F1 (%)	73.48±2.59	94.29±0.98	94.13±1.09	94.12±1.78	94.13±1.11	94.22±2.12	92.64±3.16	96.08±1.06	95.77±1.22	93.90±2.58	**97.41±0.72**

differences and spectral similarities often confound these regions, making dedicated metrics necessary

$$\text{B-OA} = \frac{|\Omega_{\text{correct}}|}{|\Omega_{\text{valid}}|} \tag{26}$$

where Ω_{valid} denotes the set of valid border pixels, Ω_{correct} denotes the subset of Ω_{valid} that are assigned the correct predicted category, and $|\cdot|$ represents the cardinality of a set. B-OA focuses on valid border pixels defined by 4-pixel-wide mask stripes. A value closer to 1 indicates superior boundary performance.

Table VI and Fig. 11 demonstrate the individual and synergistic contributions of each module. The baseline (W/O MNS&TFAB) yields the lowest metrics across all datasets. Adding only TFAB (W/O MNS) consistently improves accuracy. For example, PU OA increases from 95.97% to 96.51%. Introducing MNS alone (W/O TFAB) produces a significant performance leap in boundary regions. IP B-OA improves from 89.50% to 91.88%, proving that MNS effectively selects critical boundary nodes. The complete TFAB-DGN model achieves the best performance. It reaches 97.91% OA and 92.24% B-OA on the LK dataset.

Qualitative results in Fig. 11 provide visual evidence. Green pixels represent correct classifications, while red indicates errors. The baseline shows heavy misclassification at boundaries marked by black boxes. MNS reduces salt-and-pepper noise within classes. TFAB significantly improves boundary contours and reduces misclassified pixels. The full model maintains high within-class purity and generates the most spatially consistent classification maps. These results confirm that topological information is vital for accurate boundary discrimination.

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TABLE IV
ACCURACY COMPARISONS ON THE SA DATASET WITH 30 TRAINING SAMPLES. BOLD: BEST; UNDERLINE: SECOND-BEST

Methods	CNN	Transformer	GCN								
Class	3DCNN	DBMST	SPGCN	OGU	SPGAU	LPDGCN	GiGCN	NESSGGCN	HCGN	TFAP-GUN	Proposed
1	93.15±4.64	98.96±1.01	98.47±1.47	96.26±2.69	98.59±1.26	98.26±1.61	99.73±0.24	**99.95±0.05**	99.64±0.35	96.76±0.96	**100.00±0.00**
2	97.98±3.41	98.86±1.02	99.64±0.40	97.49±1.96	99.56±0.39	96.34±2.25	99.97±0.04	**100.00±0.00**	**100.00±0.00**	95.18±0.97	99.85±0.22
3	97.78±2.71	98.98±0.87	99.08±0.54	95.63±1.89	99.16±0.88	96.65±2.28	99.97±0.04	98.47±2.14	**100.00±0.00**	95.24±2.23	99.58±0.55
4	99.06±0.51	97.61±1.11	98.55±0.75	97.91±0.26	98.89±0.99	95.83±1.47	99.59±0.19	**99.93±0.07**	99.21±0.75	94.35±3.45	99.91±0.12
5	96.51±1.84	98.61±1.09	98.78±0.88	95.86±1.75	97.33±1.23	96.57±2.43	95.89±1.36	94.44±4.56	97.85±0.67	93.31±5.33	**98.88±0.71**
6	98.44±0.86	98.93±0.99	99.94±0.07	95.19±1.81	96.07±1.26	99.37±0.37	99.84±0.09	**100.00±0.00**	99.74±0.32	88.47±8.04	99.99±0.02
7	98.43±0.51	98.77±0.86	99.55±0.29	95.97±1.08	99.65±0.44	98.42±1.34	99.72±0.26	97.25±1.15	99.84±0.11	97.88±2.94	**99.94±0.12**
8	70.67±12.86	98.40±0.78	92.47±6.42	94.71±1.86	97.76±0.98	97.58±1.64	92.18±0.80	89.85±5.56	**98.72±0.45**	96.88±1.49	98.52±1.16
9	98.68±0.56	99.79±0.22	99.18±1.22	91.71±1.69	96.85±0.48	92.85±3.06	99.80±0.20	99.90±0.08	**100.00±0.00**	97.76±2.89	99.94±0.12
10	86.69±3.34	**99.42±0.55**	97.08±1.60	93.94±2.13	94.56±2.48	95.13±3.44	95.42±0.53	98.10±3.17	98.42±1.35	97.07±1.24	97.56±1.39
11	94.57±1.32	98.26±0.86	94.36±4.11	92.09±3.89	98.78±1.09	94.64±2.53	99.77±0.22	98.20±1.26	99.02±0.48	92.10±3.93	**99.95±0.07**
12	98.06±0.02	99.53±0.44	99.76±0.50	93.91±2.63	99.32±0.66	92.33±3.76	99.66±0.21	95.55±2.74	**99.98±0.02**	91.09±4.49	99.97±0.04
13	97.85±0.73	99.58±0.39	99.46±0.31	94.77±1.21	91.97±3.89	98.18±1.18	99.50±0.43	**100.00±0.00**	99.36±0.46	86.23±3.36	99.95±0.07
14	94.33±1.49	98.37±0.59	98.31±0.59	91.14±1.34	**99.71±0.33**	98.59±1.12	99.17±0.87	98.59±1.12	97.63±1.26	92.98±2.29	98.63±1.11
15	70.01±33.94	96.12±2.11	87.79±8.32	92.64±2.21	92.29±2.56	97.90±1.44	93.06±0.65	83.06±7.67	**98.65±1.23**	92.23±1.01	97.42±1.74
16	84.57±3.68	98.93±1.08	96.14±2.40	95.86±2.10	97.96±1.78	96.40±1.89	99.18±0.31	99.16±1.31	99.13±0.54	98.64±0.28	**99.27±1.78**
OA (%)	82.53±4.55	97.97±1.06	96.12±0.87	95.68±1.19	97.48±0.46	96.40±1.41	96.73±0.23	97.89±1.65	99.11±0.35	95.62±1.29	**99.18±0.26**
AA (%)	90.92±1.53	98.18±0.88	96.41±0.47	95.73±1.13	98.38±0.13	95.34±1.81	98.30±0.09	98.39±1.12	99.22±0.49	94.63±0.78	**99.33±0.12**
Kappa (%)	80.71±4.89	97.72±0.96	95.53±0.97	95.64±1.21	97.70±0.19	94.03±1.62	96.45±0.25	96.32±2.34	98.79±0.48	94.82±0.45	**98.94±0.31**
F1 (%)	78.29±1.28	97.64±1.28	95.12±1.19	94.29±1.43	97.59±0.48	95.89±1.23	95.18±1.29	98.28±0.18	98.95±0.87	93.74±1.78	**99.15±0.21**

TABLE V
ACCURACY COMPARISONS ON THE LK DATASET WITH 30 TRAINING SAMPLES. BOLD: BEST; UNDERLINE: SECOND-BEST

Methods	CNN	Transformer	GCN								
Class	3DCNN	DBMST	SPGCN	OGU	SPGAU	LPDGCN	GiGCN	NESSGGCN	HCGN	TFAP-GUN	Proposed
1	96.16±2.19	99.69±0.32	99.54±0.66	96.50±2.13	98.59±1.26	99.21±0.77	97.11±2.18	**99.94±0.06**	99.89±0.09	96.68±2.16	97.70±1.25
2	53.91±7.14	99.12±0.87	99.28±0.72	99.60±0.41	99.54±0.87	99.16±0.72	99.19±0.67	99.87±0.24	96.71±2.16	**99.92±0.07**	99.44±0.55
3	88.72±2.19	99.68±0.44	98.64±1.06	99.93±0.08	95.89±2.17	**100.00±0.00**	96.51±3.19	**100.00±0.00**	92.21±3.16	87.23±4.16	**100.00±0.00**
4	87.03±5.82	95.25±1.23	91.97±2.09	93.24±1.59	98.59±1.07	93.09±1.69	90.81±5.23	94.76±5.77	**99.50±0.48**	95.17±2.75	97.74±1.17
5	52.57±7.95	99.65±0.55	99.06±0.88	99.29±0.77	99.16±0.77	99.17±0.83	99.85±0.16	99.18±0.79	91.41±3.17	99.66±0.34	**99.92±0.09**
6	90.70±3.11	99.85±1.09	99.48±0.56	**100.00±0.00**	99.92±0.09	**100.00±0.00**	99.64±0.33	95.56±3.18	97.93±1.67	99.91±0.09	**100.00±0.00**
7	99.40±0.52	99.88±0.10	99.89±0.16	99.91±0.09	98.68±1.12	99.91±0.10	99.75±0.20	99.91±0.08	99.95±0.06	99.91±0.10	**99.97±0.03**
8	71.04±4.22	86.64±3.29	86.18±3.59	92.21±1.99	96.87±1.23	94.47±2.19	82.74±5.26	**97.36±1.65**	89.81±3.89	89.25±3.73	96.99±1.29
9	76.53±4.31	89.44±1.98	89.82±5.19	91.51±4.19	88.19±3.74	97.69±1.20	70.71±12.66	77.12±8.89	83.06±6.49	89.55±2.76	**98.92±1.67**
OA (%)	89.73±2.59	97.28±0.59	96.77±1.48	96.89±1.11	96.09±1.19	97.66±1.23	96.51±2.15	97.65±0.81	97.59±0.73	97.08±1.09	**97.91±0.32**
AA (%)	79.56±2.46	96.19±0.98	96.22±1.26	96.21±1.28	96.16±1.26	96.19±1.44	94.48±1.30	96.85±2.16	96.58±1.92	95.25±2.18	**97.53±0.65**
Kappa (%)	86.37±3.75	96.14±1.28	95.61±1.37	95.87±2.09	96.38±1.30	96.78±1.07	94.97±1.19	96.17±2.14	97.19±1.29	96.17±2.29	**97.31±0.40**
F1 (%)	82.69±2.96	94.26±2.09	92.99±2.49	92.79±1.86	95.98±1.89	95.96±1.01	94.19±1.76	95.34±2.17	96.18±1.48	93.96±2.51	**96.73±0.78**

D. Validity of Topological Information

1) Magnitude Score Top-k Boundary Nodes Analysis: Fig. 12 visualizes the Top-k boundary nodes selected by Magnitude scores to support Contribution 2. These nodes are automatically identified during testing and marked by red asterisks. We set k to 100 for IP, 200 for PU, 150 for SA, and 50 for LK. Selected nodes are concentrated at junctions between categories, turning points in contours, and complex geometric structures rather than being randomly distributed. In the IP dataset, nodes appear primarily in irregular background areas. The PU model focuses on scattered boundary regions. For SA and LK, nodes cluster densely along the interfaces between adjacent regions. Magnitude weights effectively capture key transitions in the topological structure. These selected nodes correspond to areas most prone to misclassification. Our strategy leverages topological importance to identify discriminative features that traditional methods often miss.

2) Feature Separability in PH-Weighted Boundary Pixels: We conducted a separability experiment on pixels within boundary nodes to support Contribution 3. Fig. 13 compares Euclidean-weighted edges based on spectral information with our PH-weighted edges. Silhouette coefficient (SC) measures cluster separation as a percentage. The Calinski–Harabasz (CH) index reflects clustering performance through variance ratios. Higher values for both indicate better intraclass compactness and interclass separation. Despite limited boundary samples in the IP, PU, and LK datasets, the PH-weighted GCN (red border) shows markedly better separation. The spectrally weighted GCN (black border) exhibits severe feature confusion in the SA dataset. PH weighting effectively

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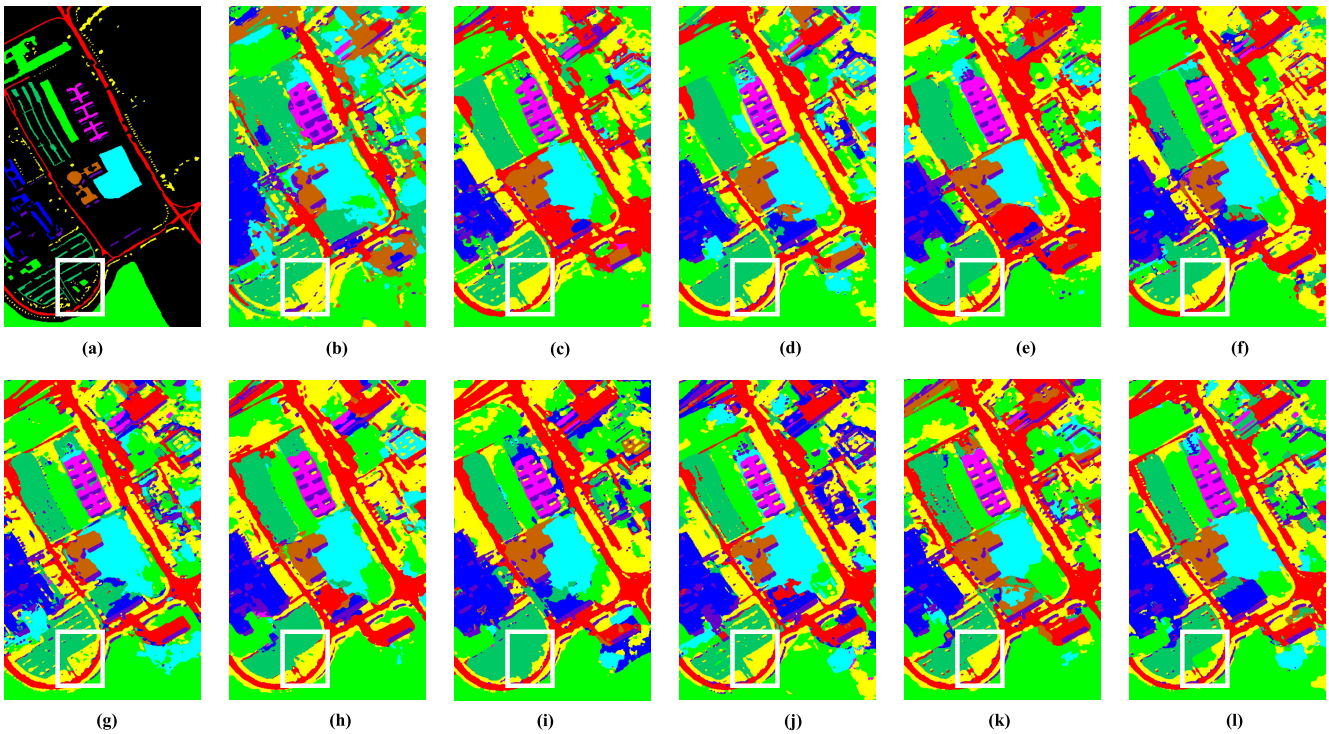


Fig. 8. Classification maps of different methods on the PU dataset. (a) Ground truth. (b) 3DCNN. (c) DBMST. (d) SPGCN. (e) OGU. (f) SPGAU. (g) LPDGCN. (h) GiGCN. (i) NESSGGCN. (j) HCGN. (k) TFAP-GUN. (l) Proposed.

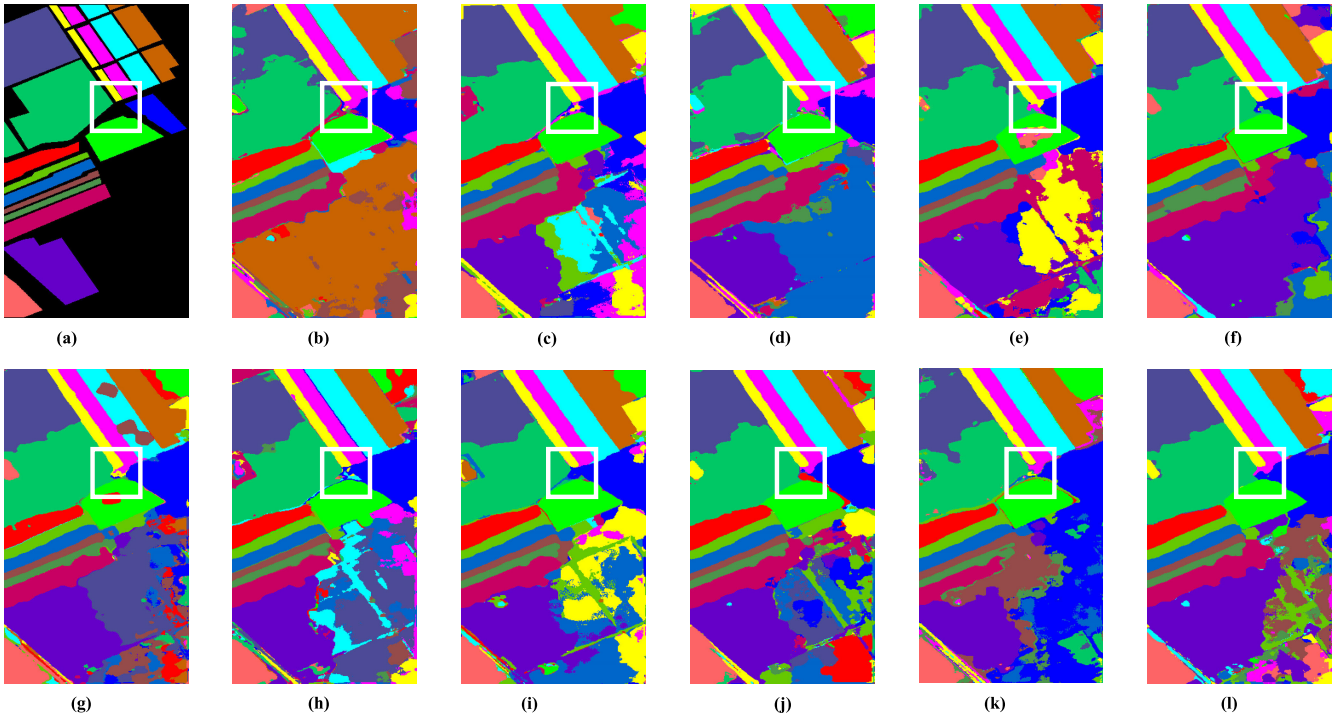


Fig. 9. Classification maps of different methods on the SA dataset. (a) Ground truth. (b) 3DCNN. (c) DBMST. (d) SPGCN. (e) OGU. (f) SPGAU. (g) LPDGCN. (h) GiGCN. (i) NESSGGCN. (j) HCGN. (k) TFAP-GUN. (l) Proposed.

alleviates this confusion. Quantitatively, IP and PU datasets show the most pronounced improvements in SC and CH metrics. TFAB adaptively adjusts unreliable edge weights and optimizes message passing through stable topological signals.

3) Effectiveness in Controlled Boundary Scenarios: We constructed challenging training scenarios to evaluate the value of topological information in TFAB-DGN. The experiments varied the proportion of boundary pixels within a fixed set of 30 labeled samples per class. Table VII shows that model

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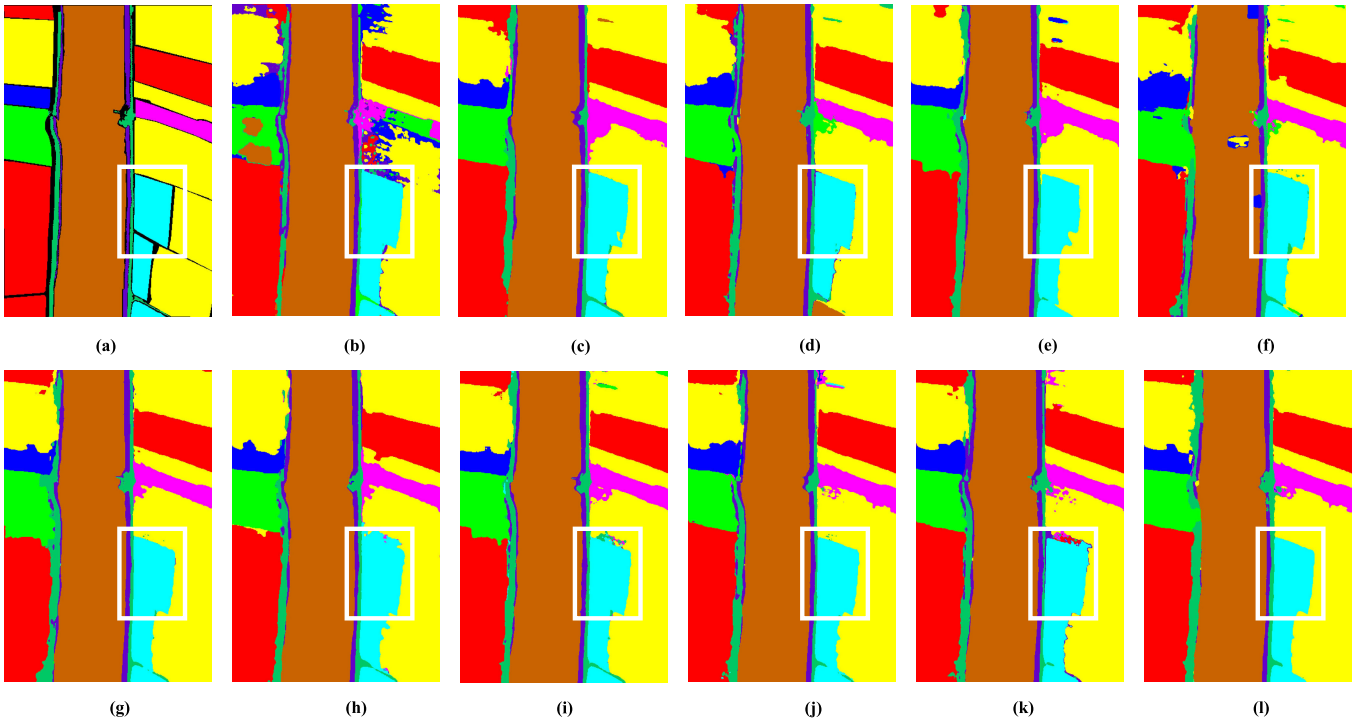


Fig. 10. Classification maps of different methods on the LK dataset. (a) Ground truth. (b) 3DCNN. (c) DBMST. (d) SPGCN. (e) OGU. (f) SPGAU. (g) LPDGCN. (h) GiGCN. (i) NESSGGCN. (j) HCGN. (k) TFAP-GUN. (l) Proposed.

TABLE VI

ABLATION STUDIES FOR THE PROPOSED METHOD. BOLD: BEST; ASTERISK: BOUNDARY-OA

Datasets	Methods	OA	AA	Kappa	F1	B-OA*
IP	W/O MNS&TFAB	93.12±1.16	96.09±1.09	91.98±1.34	92.12±1.09	89.50±0.89
	W/O MNS	93.91±1.28	96.71±1.03	92.68±1.62	92.51±1.32	90.53±0.67
	W/O TFAB	94.74±1.01	97.17±0.42	93.98±1.15	92.96±2.14	91.88±0.77
	Proposed	**95.16±0.52**	**97.39±0.82**	**94.46±1.02**	**93.17±1.71**	**92.80±0.59**
PU	W/O MNS&TFAB	95.97±1.65	97.47±1.23	94.70±1.14	95.75±1.17	93.98±0.84
	W/O MNS	96.51±1.08	96.63±1.55	95.36±1.38	96.19±1.84	94.19±0.57
	W/O TFAB	96.27±1.26	97.95±0.48	95.16±1.35	96.35±1.39	94.64±0.96
	Proposed	**97.65±0.39**	**98.35±0.24**	**96.82±0.58**	**97.41±0.72**	**97.25±0.88**
SA	W/O MNS&TFAB	97.19±1.00	97.01±1.23	95.99±1.20	96.55±1.37	94.27±1.07
	W/O MNS	97.86±1.16	97.13±0.95	95.73±1.29	96.79±1.01	94.50±0.74
	W/O TFAB	98.85±0.26	99.25±0.09	98.72±0.29	98.92±0.18	96.47±0.58
	Proposed	**99.18±0.26**	**99.33±0.12**	**98.94±0.31**	**99.15±0.21**	**97.99±0.21**
LK	W/O MNS&TFAB	96.27±0.39	93.56±1.29	95.13±1.09	93.78±1.41	89.93±0.78
	W/O MNS	97.36±0.74	96.72±2.01	96.54±0.96	95.08±1.73	90.34±0.44
	W/O TFAB	97.05±0.96	97.47±0.47	96.15±1.24	95.79±1.21	90.39±0.77
	Proposed	**97.91±0.32**	**97.53±0.65**	**97.31±0.40**	**96.73±0.78**	**92.24±0.94**

performance improves as the proportion of boundary pixels increases. Both magnitude and PH contribute significant value to boundary classification. Optimal performance occurs when the boundary pixel proportion reaches 0.7–0.9. This demonstrates the strong adaptability of TFAB-DGN to complex boundary environments.

E. Hyperparameter Analysis

1) Impact of the Top-k and Hidden Layer Dimension: Fig. 14 provides a 3-D heatmap of OA across different combinations of boundary node selection (Top-k) and hidden layer dimensions. Excessive Top-k values introduce nondiscriminative features and increase computational complexity. Large node counts can also disrupt the spatial–spectral consistency of smooth regions. Hidden layer dimensions greater than 60 increase overfitting risks and degrade performance. Optimal results occur across all datasets when Top-k is 3 or 5, and the hidden dimension is 40 or 50. This configuration strikes the best balance between model complexity and effective information capture.

2) Impact of the Number of Training Samples: Fig. 15 illustrates OA trends as training samples per class vary from 5 to 30. All methods improve with more labeled data, yet TFAB-DGN maintains a significant lead throughout the range. The advantage is especially clear with extremely scarce data, like 5 or 10 samples per class. Our model consistently outperforms benchmarks such as 3DCNN, SPGCN, and GiGCN across all datasets. Topology-aware boundary features allow the model to learn more effectively from small sample sets.

3) Impact of the Segmentation Scale: Fig. 16 shows OA as a function of the segmentation scale δ ranging from 40 to 500. SA (green curve) demonstrates high robustness with minimal fluctuations across the entire range. IP and LK show more sensitivity to scale changes. IP peaks at $\delta = 60$ with accuracy exceeding 95%. LK peaks at $\delta = 450$ and approaches 98%. TFAB-DGN does not require an extremely precise scale parameter. The optimal value is easily selected in practical applications. These results help balance the preservation of object details with computational complexity.

F. Running Time and Model Complexity

Table VIII compares training time, testing time, total runtime, and parameter counts across four datasets. TFAB-DGN

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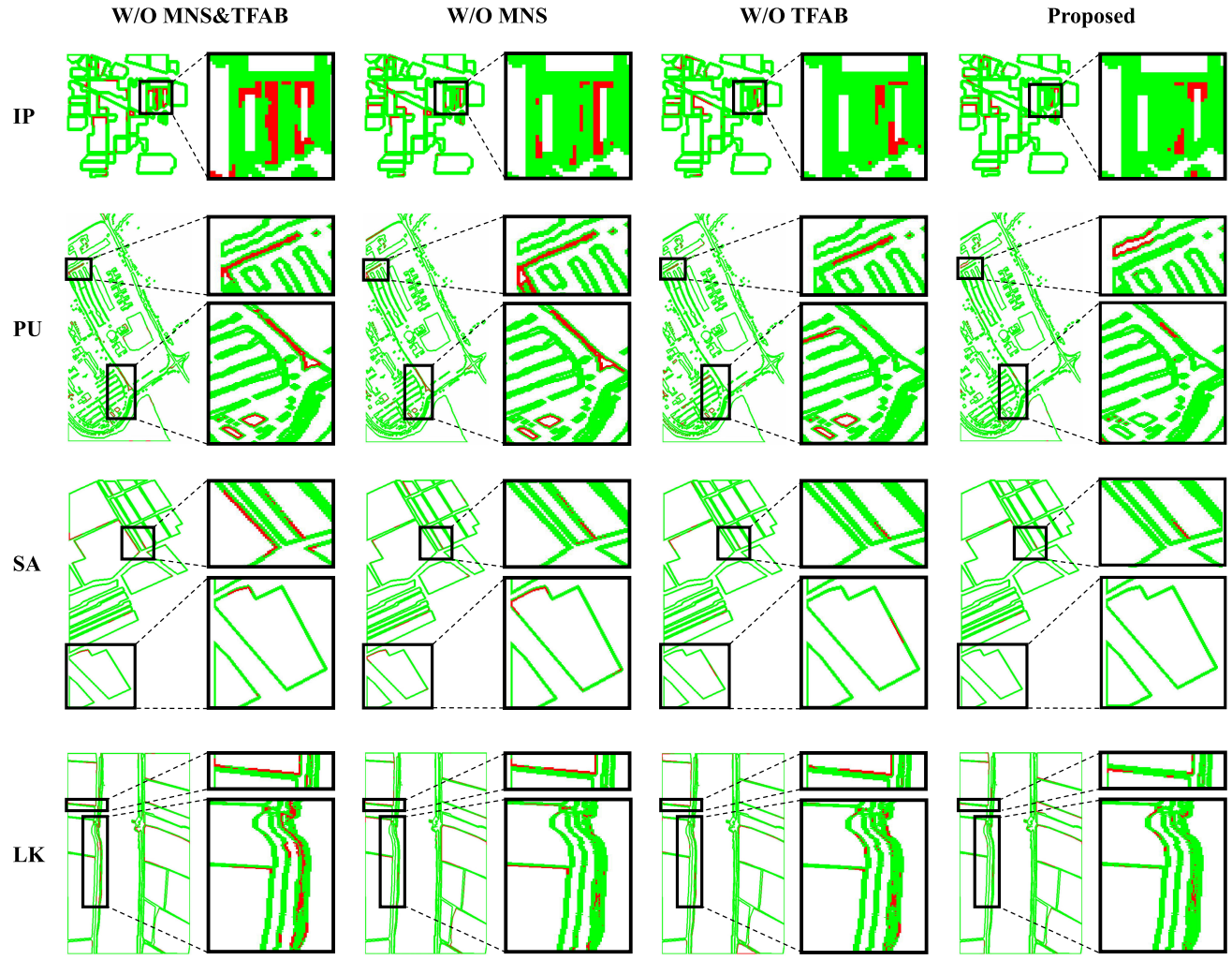


Fig. 11. Ablation studies for the proposed method. Green: correct classification, Red: incorrect classification, Outline: significant boundary change.

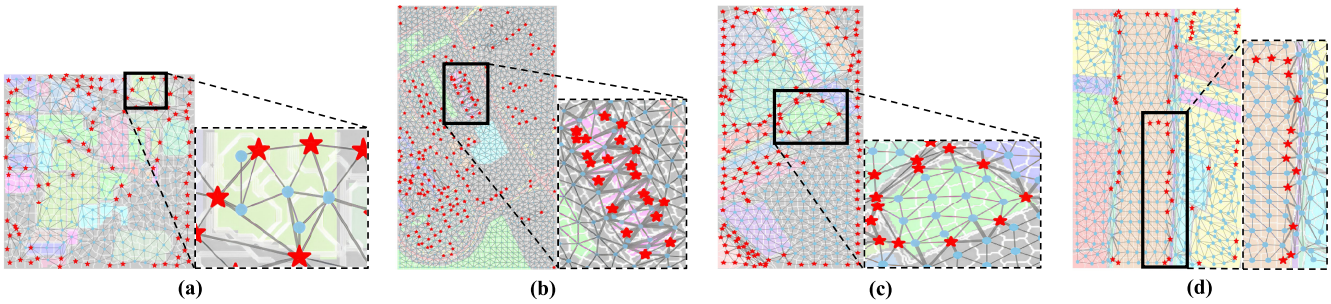


Fig. 12. Magnitude score Top-k boundary nodes visualization. Star: Top-k boundary nodes, blue dots: smooth nodes. (a) IP. (b) PU. (c) SA. (d) LK.

balances model complexity and inference efficiency. Parameter counts for IP, PU, SA, and LK are 104.328K, 98.927K, 104.536K, and 72.651K, respectively. These values are significantly lower than those of GiGCN, NESSGGCN, and HCGN. Our dual-graph framework remains lightweight through effective structural design. While training and total runtimes are not the fastest, testing time remains at a moderate level. On the LK dataset, testing takes only 1.72 s, outperforming other models. This advantage suits practical scenarios with limited computational resources or rapid testing requirements.

Offline precomputation of magnitude-based weight maps is recorded in Table IX. This one-time step ranges from 33.64 to 278.69 s per dataset. IP shows the highest efficiency. LK and PU require more time due to their larger image sizes and spatial complexity. This precomputation occurs only during preprocessing to provide stable node-importance priors

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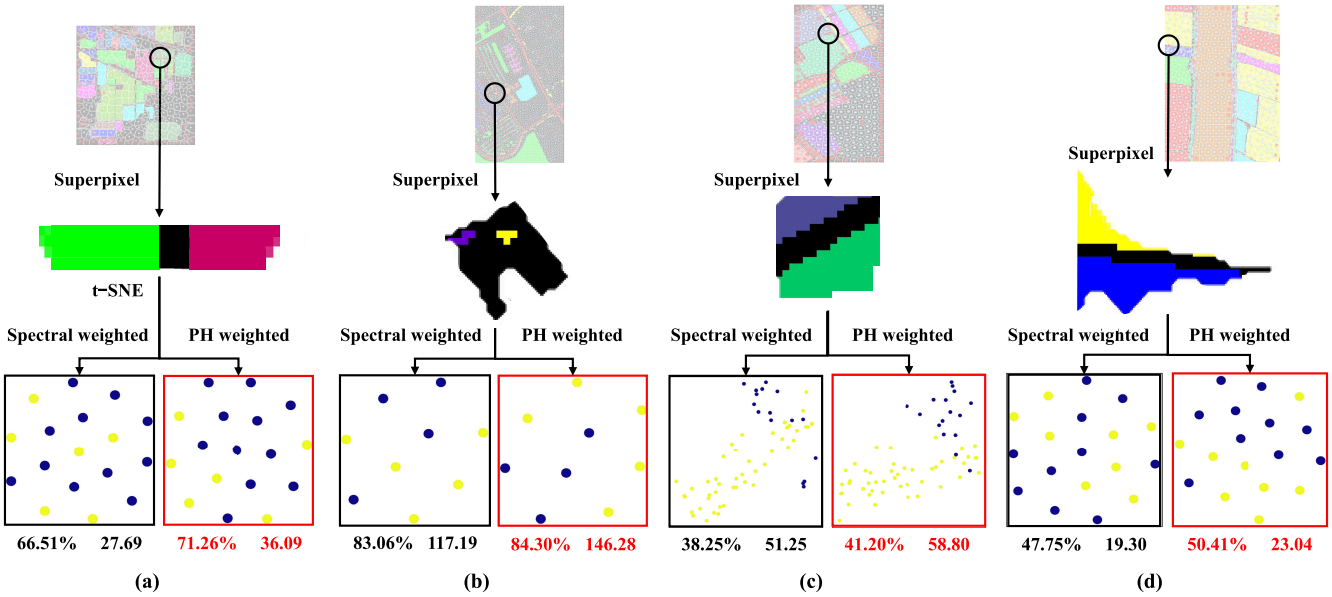


Fig. 13. Feature separability in PH-weighted boundary pixels. Star: Top-k boundary superpixel, black border: spectral weighted, red border: PH-weighted. The t-SNE clustering performance: (left) SC and (right) CH. A larger value indicates that the extracted features are better clustered. (a) IP. (b) PU. (c) SA. (d) LK.

TABLE VII

BOUNDARY PIXEL TRAINING SAMPLES WITH CONTROLLED RATIOS OF CLASSIFICATION PERFORMANCE COMPARISON. BOLD: BEST

Boundary pixel training ratio	IP				PU				SA				LK			
	OA	AA	Kappa	F1	OA	AA	Kappa	F1	OA	AA	Kappa	F1	OA	AA	Kappa	F1
0.0	93.78	95.25	92.89	89.60	93.93	96.31	92.75	93.51	98.97	99.33	98.85	99.00	96.83	96.65	95.85	94.93
0.2	94.14	95.11	93.01	90.47	95.49	97.20	93.81	94.71	98.64	99.41	98.49	99.34	97.27	97.45	96.43	95.31
0.4	94.92	96.25	93.95	91.67	95.20	96.37	93.44	94.87	97.42	98.98	97.13	98.68	97.08	97.52	96.91	95.12
0.5	95.31	96.59	94.72	95.25	96.39	97.64	95.25	96.47	99.06	99.52	98.96	99.52	97.01	97.85	**97.79**	96.09
0.7	**96.26**	98.04	**95.72**	**96.81**	97.30	97.86	96.43	97.11	**99.80**	**99.70**	**99.77**	**99.71**	97.84	**97.96**	97.17	**96.92**
0.9	96.04	**98.48**	95.33	95.45	**98.83**	**98.78**	**98.45**	**98.58**	99.74	99.68	99.71	99.70	**98.27**	97.88	97.43	96.74
Random training	95.16	97.39	94.46	93.17	97.65	98.35	96.82	97.41	99.18	99.33	98.94	99.15	97.91	97.53	97.31	96.73

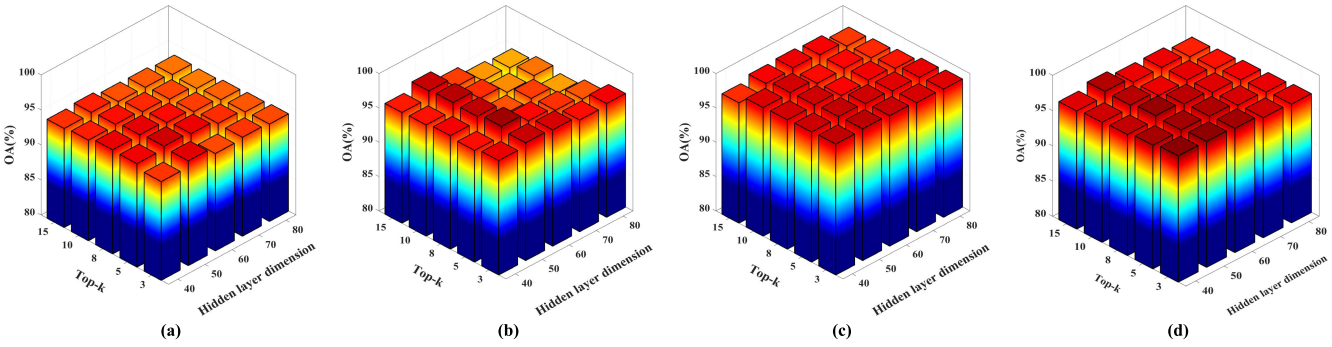


Fig. 14. Impact of different Top-k and dimension of the hidden layers on the model across four datasets. (a) IP. (b) PU. (c) SA. (d) LK.

for training. The magnitude-based weighting strategy proves efficient and practical for real-world applications.

V. LIMITATIONS AND FUTURE WORK

TFAB-DGN effectively classifies HSI boundaries but faces several limitations. High computational complexity remains a primary challenge for large-scale data. PH calculations are time-consuming and hinder efficiency in practical applications. The model also shows sensitivity to hyperparameters like superpixel scale δ and Top-k selection. Optimal ranges exist, yet manual adjustments are still necessary for specific scenarios. This indicates a lack of full parameter adaptability.

Future research should improve PH efficiency using approximation algorithms or parallel computing. Knowledge

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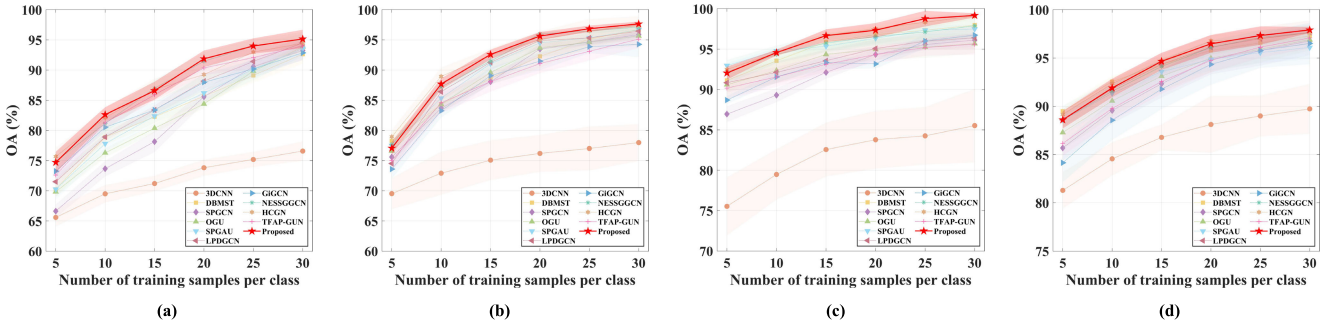


Fig. 15. Impact of different training sample numbers on the model across four datasets. (a) IP. (b) PU. (c) SA. (d) LK.

TABLE VIII

COMPARISON OF TRAINING TIME, TESTING TIME, RUNNING TIME, AND NUMBER OF PARAMETERS FOR THE COMPARED METHODS ON THE FOUR CONSIDERED DATASETS. BOLD: BEST; UNDERLINE: SECOND-BEST

Dataset	Time/Complexity	3DCNN	DBMST	SPGCN	OGU	SPGAU	LPDGCN	GiGCN	NESSGGCN	HCGN	TFAP-GUN	Proposed
IP	Train(s)	22.64	137.40	1117.67	13.89	48.32	160.13	34.37	26.78	15.84	**7.55**	250.20
	Test(s)	0.56	5.29	44.34	0.38	2.19	1.96	1.07	1.05	0.13	**0.01**	1.04
	Run(s)	23.20	142.69	1162.01	14.27	50.51	162.09	35.44	27.83	15.97	**7.56**	251.24
	Params(K)	46.110	216.560	**22.61**	143.820	102.422	142.420	269.300	313.964	452.000	122.920	104.328
PU	Train(s)	12.24	130.20	672.81	575.26	69.56	743.26	62.56	56.48	47.33	**8.70**	590.56
	Test(s)	3.66	9.85	143.24	0.95	7.35	12.08	2.84	11.25	0.63	**0.62**	3.00
	Run(s)	15.90	140.05	816.05	576.21	76.91	755.34	65.40	67.73	47.96	**9.32**	593.56
	Params(K)	35.530	187.850	**20.550**	84.780	89.357	100.060	640.55	384.620	453.000	78.850	98.927
SA	Train(s)	638.79	726.68	2171.68	119.40	291.95	717.47	75.96	55.84	105.54	**17.15**	683.22
	Test(s)	23.59	10.95	216.07	0.13	24.31	19.34	5.79	10.48	**0.08**	3.02	3.50
	Run(s)	662.38	737.63	2387.75	119.53	316.26	736.81	81.75	66.32	105.62	**20.17**	686.72
	Params(K)	52.540	188.430	**22.86**	53.030	102.942	142.420	660.550	374.520	438.000	124.880	104.536
LK	Train(s)	21.45	97.20	1289.52	110.78	262.81	273.81	91.42	58.71	120.71	**10.86**	319.57
	Test(s)	15.35	3.43	58.12	6.57	13.98	9.38	15.28	16.84	**1.20**	1.75	1.72
	Run(s)	36.80	100.63	1347.64	117.35	276.79	283.19	106.70	75.55	121.91	**12.61**	321.39
	Params(K)	44.270	150.980	**25.770**	91.517	111.912	101.180	682.540	386.040	460.000	86.296	72.651

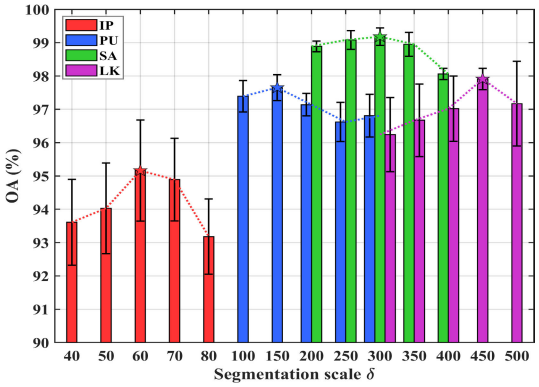


Fig. 16. Impact of different superpixel segmentation scales on OA was analyzed across four datasets.

TABLE IX

RUNNING TIME OF OFFLINE PRECOMPUTATION FOR THE MAGNITUDE-BASED WEIGHT MAPS ON THE FOUR DATASETS

Dataset	IP	PU	SA	LK
Run(s)	33.64	238.16	139.87	278.69

distillation [67] and more streamlined extraction methods offer promising alternatives. Developing flexible adaptive mechanisms could reduce reliance on fixed superpixel segmentation. The topological awareness framework can also be extended to other tasks. These include cross-domain learning [68], change detection [69], and contrastive learning [70]. Such extensions will further validate the generalization capability of the model.

VI. CONCLUSION

TFAB-DGN addresses insufficient boundary discriminability in HSI classification by integrating magnitude and PH into GCNs. Local magnitude quantifies HSI structures to identify spectral–spatial inconsistencies at boundaries. PH extracts stable topological features to guide information propagation through a reweighted adjacency matrix. This process significantly enhances the discriminative power of boundary nodes. Evaluation across four datasets demonstrates superior performance after multiscale feature fusion. Ablation experiments and visualizations confirm the specific contributions of each module at HSI boundaries. This research offers new insights for HSI analysis. Future efforts will focus on more efficient and stable methods for topological information mining.

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